The Sachdev–Ye–Kitaev model

Lecture notes for mathematicians

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Abstract

The Sachdev–Ye–Kitaev model is a Gaussian random Hamiltonian generated by a Clifford algebra. These notes introduce its finite-dimensional formulation and the large-N equations for its thermal correlation functions. We develop the fermionic algebra, explain the bilocal description, and derive the conformal scaling predicted by the infrared equations. We then discuss the Schwarzian effective action, four-point functions, quantum chaos, and the relation to nearly-AdS_2 gravity. The intended reader knows probability and linear algebra; quantum-mechanical terminology is defined as it appears. Finite-N results are proved where elementary; diagrammatic and effective-theory arguments are accompanied by their assumptions. The main physics reference is Rosenhaus [1].

Contents

1 The finite-dimensional model 3
- 1.1 Introduction . . . 3
- 1.2 Fermions and Majorana operators . . 4
- 1.3 Grassmann algebra and correlators . 8
- 1.4 The SYK Hamiltonian . . . 10

2 Large-N and infrared theory 12
- 2.1 The melonic large-N limit . . . 12
- 2.2 The bilocal collective action . . . 13
- 2.3 The infrared conformal limit . . . 14
- 2.4 The Schwarzian soft mode . . . 16
- 2.5 The four-point ladder kernel . . . 20

3 Chaos and quasiparticles 21
- 3.1 Quantum chaos . . . 21
- 3.2 Fermi liquids . . . 23
- 3.3 SYK and non-Fermi-liquid behavior . 24
- 3.4 Transport and spatial extensions . . . 25
- 3.5 Random-matrix statistics . . . 26

4 Nearly-AdS2 gravity 28
- 4.1 Geometric preliminaries . . . 28
- 4.2 JT gravity and the Schwarzian . . . 29
- 4.3 Black-hole thermodynamics . . . 30
- 4.4 The holographic dictionary . . . 30
- 4.5 Black holes and chaos . . . 31
- 4.6 Complex SYK and charge . . . 32

5 Probability and further questions 32
- 5.1 The probabilistic formulation . . . 32
- 5.2 Scope of the available results . . . 35
- 5.3 Further questions . . . 35

A Supplementary calculations 38
- A.1 Collective-field calculations . . . 38
- A.2 The infrared Fourier transform . . . 39
- A.3 The Clifford trace identity . . . 40
- A.4 Schwarzian invariance . . . 40
- A.5 Euclidean and real time . . . 40
- A.6 Fock, Clifford, and Gaussian identities 41
- A.7 Thermal trace calculations . . . 42
- A.8 Conformal distributions and coordinate changes . . . 44
- A.9 What the formal linearizations prove 46
- A.10 Antiunitary symmetries and the late-time average . . . 47
- A.11 Geometric computations . . . 47
- A.12 Comparison of conventions . . . 49

B Terminology 50 References 51

1. The model and its finite-dimensional formulation

1.1. Introduction

Definition 1.1 (Quantum-mechanical vocabulary). A finite quantum system consists of a complex Hilbert space $\mathcal{H}$ and a self-adjoint Hamiltonian H, the operator specifying its energy. A pure state is a unit vector modulo multiplication by a complex number of modulus one. A mixed state is a positive semidefinite matrix ρ with $\operatorname{Tr}\rho = 1$. An observable is a self-adjoint matrix A, with expectation $\operatorname{Tr}(\rho A)$. We write $A^* = A^\dagger$, $[A, B] = AB - BA$, and $\{A, B\} = AB + BA$. A ket $|v\rangle$ denotes a vector, a bra $\langle v|$ its adjoint, and $\langle v|A|w\rangle$ a matrix entry.

An energy eigenvalue E satisfies $Hv = Ev$. Real-time evolution is the unitary group $e^{-itH/\hbar}$; in the Heisenberg convention the state is fixed and $A(t) = e^{itH/\hbar}Ae^{-itH/\hbar}$. A many-body system has a Hilbert space built from many elementary modes. Here a mode is one fermionic degree of freedom and its flavor is its label i. All-to-all means that every subset of q distinct labels is allowed to interact. There is no distance between labels and no spatial coordinate.

Temperature $T > 0$ enters through $\beta = (k_B T)^{-1}$, where k_B converts temperature to energy. Planck's constant $\hbar$ converts inverse energy to time. Setting both constants to one gives $\beta = 1/T$. Disorder means randomness in the couplings of a Hamiltonian sampled before its dynamics is computed; it is distinct from the probabilistic interpretation of a quantum state.

The Sachdev-Ye-Kitaev model is a random quantum many-body system with no spatial coordinate. In its most common form it consists of N Majorana fermions and random all-to-all quartic interactions. For even interaction order q, the Hamiltonian is

$$H_q = \sigma_N \sum_{I \in \mathcal{I}_N} g_I \psi_I, \qquad \{\psi_i, \psi_j\} = 2\delta_{ij} I, \tag{1.1}$$

where N is even, $2 \le q \le N$ is even,

$$\mathcal{I}_N = \{(i_1, \dots, i_q) : 1 \le i_1 < \cdots < i_q \le N\}, \qquad g_I \overset{\text{iid}}{\sim} \mathcal{N}(0, 1), \tag{1.2}$$

and

$$\psi_I := i^{q/2} \psi_{i_1} \cdots \psi_{i_q}, \qquad \sigma_N^2 = \frac{(q-1)!}{N^{q-1}}.$$

At finite N, this defines a random self-adjoint operator on a Hilbert space of dimension $2^{N/2}$.

The main question is how the thermal observables of this random matrix behave as N tends to infinity. The matrix dimension grows exponentially, whereas the number of independent random coefficients grows polynomially for fixed q. The resulting correlations between matrix entries distinguish SYK from the usual ensembles of random matrix theory.

For $q = 2$, a change of one-particle basis diagonalizes the Hamiltonian. For fixed $q \ge 4$, the large-N calculation instead leads to a nonlinear equation for a two-time correlation function. Its low-frequency approximation has a power-law solution. Fluctuations around that solution lead to the Schwarzian action, which also arises at the boundary of nearly-AdS_2 gravity. Our purpose is to explain these reductions and the assumptions needed at each passage.

The organization is as follows. Section 1 develops the finite-dimensional model. Section 2 introduces the large-N equations and their infrared approximation. Sections 3 and 4 discuss the physical interpretations. Section 5 returns to the probabilistic formulation and outstanding questions.

The appendices supply algebraic calculations and a terminology guide. A first reading may stop after the Schwarzian action and resume with the probabilistic discussion.

Unless stated otherwise, q is fixed and N tends to infinity through even integers. Statements about the interacting infrared regime concern $q \geq 4$, usually $q = 4$. We use $\hbar = k_B = 1$ and set the interaction scale to one. The symbols $\mathbb{E}$ and $\langle\cdot\rangle_{\beta,g}$ denote disorder expectation and thermal expectation at fixed disorder, respectively. We write $a_N \asymp b_N$ when their positive ratio stays between two constants independent of N.

The mean-field analogy. For orientation, the Curie–Weiss energy of spins $s_i \in \{-1, 1\}$ is $-J(\sum_i s_i)^2/(2N)$, and its magnetization is $m = N^{-1}\sum_i s_i$. The Sherrington–Kirkpatrick energy is $-N^{-1/2}\sum_{i<j} g_{ij}s_is_j$. The overlap of two spin configurations is $R(s, s') = N^{-1}\sum_i s_is'_i$. A mean-field description replaces many microscopic coordinates by a self-consistent macroscopic observable, called an order parameter. These examples explain the terminology; their thermodynamic limit theorems are not needed here.

In the Curie-Weiss model, one starts from N spins and obtains in the thermodynamic limit a variational problem for one scalar order parameter m. In the Sherrington-Kirkpatrick model, disorder introduces more elaborate order parameters, including overlap distributions. In SYK the large-N order parameter is a two-time function $G(\tau_1, \tau_2)$. Thus the microscopic system is zero-dimensional in space, but the order parameter is infinite-dimensional because it depends on time.

The central reduction is

$$\psi_1, \ldots, \psi_N \quad \rightsquigarrow \quad (G, \Sigma),$$

where G is the Green kernel and Σ is the self-energy. The collective action is proportional to N, so $1/N$ acts as a semiclassical fluctuation parameter. Strong coupling and analytic control coexist because the small parameter is not the interaction strength but the inverse number of fermion species.

Status of the arguments. An action is a functional appearing in a weight e^{-I} in Euclidean time (or $e^{iI/\hbar}$ in real time). A saddle is a stationary point of that functional. A saddle approximation expands the action about such a point; it requires further estimates to approximate an integral. An effective action describes selected observables after other variables have been eliminated. "Semiclassical" means that fluctuations about a stationary configuration are small. Below, "Details omitted" explicitly identifies an imported result, a formal passage, or a calculation not verified here; it is not a claim of a missing elementary proof. Definitions, propositions, and their proofs concern ordinary finite-dimensional mathematics unless another setting is specified. A *formal statement* records a conclusion of diagrammatic or saddle-point analysis without claiming a uniform error estimate. An *effective description* concerns specified observables in a scaling regime; equality of effective actions alone does not identify microscopic Hilbert spaces. We use these distinctions throughout, especially when discussing the infrared limit and gravity.

1.2. Fermions and Majorana operators

We begin with exterior algebra, which gives a concrete representation of the fermionic operators.

Antisymmetric Fock space. Let the one-particle Hilbert space be $\mathfrak{h} = \mathbb{C}^M$. The fermionic Fock space is

$$\mathcal{F}_-(\mathfrak{h}) = \bigoplus_{r=0}^{M} \Lambda^r \mathfrak{h}. \tag{1.3}$$

Its dimension is

$$\dim \mathcal{F}_-(\mathbb{C}^M) = \sum_{r=0}^{M} \binom{M}{r} = 2^M.$$

Choose an orthonormal basis $e_1, \dots, e_M$. Exterior multiplication by e_a defines the creation operator $c_a^\dagger \psi = e_a \wedge \psi$, and its adjoint c_a is the annihilation operator.

Proposition 1.2 (Canonical anticommutation relations). *The operators $c_a, c_a^\dagger$ satisfy*

$$\begin{aligned} \{c_a, c_b^\dagger\} &= \delta_{ab} I, \\ \{c_a, c_b\} &= 0, \\ \{c_a^\dagger, c_b^\dagger\} &= 0. \end{aligned}$$

Proof. On a basis wedge, the adjoint of exterior multiplication is

$$c_a(e_{j_1} \wedge \cdots \wedge e_{j_r}) = \sum_{\ell=1}^{r} (-1)^{\ell-1} \delta_{a j_\ell} e_{j_1} \wedge \cdots \wedge \widehat{e}_{j_\ell} \wedge \cdots \wedge e_{j_r}.$$

Consequently $c_a(e_b \wedge u) = \delta_{ab} u - e_b \wedge c_a u$, which proves the first relation. Exterior multiplication anticommutes; taking adjoints gives the remaining relation. □

The nilpotence $(c_a^\dagger)^2 = 0$ is the algebraic form of Pauli exclusion. The number operator $n_a = c_a^\dagger c_a$ satisfies $n_a^2 = n_a$, $\operatorname{spec}(n_a) = \{0, 1\}$. A convenient orthonormal basis is $|n_1, \dots, n_M\rangle$, $n_a \in \{0, 1\}$. Thus M complex fermionic modes have the same Hilbert-space dimension as M qubits, but the natural generators satisfy anticommutation relations rather than commuting across tensor factors.

Parity and the graded algebra. The total number operator is

$$\mathcal{N} = \sum_{a=1}^{M} n_a.$$

The fermion-parity operator is $(-1)^F = e^{i\pi\mathcal{N}}$.

Here F is conventional notation for fermion number modulo two. On $\Lambda^r \mathfrak{h}$, parity equals $(-1)^r$. An operator is even or odd when conjugation by parity multiplies it by $+1$ or -1, respectively. This is the $\mathbb{Z}/2\mathbb{Z}$ grading; a symmetry sector is an invariant eigenspace of a commuting symmetry, in particular one of the two parity eigenspaces. Operators which change particle number by an even amount commute with parity; odd operators anticommute with parity. Physical Hamiltonians are typically even. In SYK the quartic Hamiltonian is even, hence $[H, (-1)^F] = 0$. This grading is important when discussing spectral degeneracies and random-matrix symmetry classes.

Majorana fermions and Clifford algebras. For each complex mode define two self-adjoint operators

$$\psi_{2a-1} = c_a + c_a^\dagger,$$
$$\psi_{2a} = i(c_a - c_a^\dagger).$$

Then

$$\psi_i^* = \psi_i, \qquad \{\psi_i, \psi_j\} = 2\delta_{ij} I, \qquad \psi_i^2 = I. \tag{1.4}$$

Conversely,

$$c_a = \frac{\psi_{2a-1} - i\psi_{2a}}{2}.$$

Hence $N = 2M$ Majorana fermions act irreducibly on $\mathcal{H}_N \cong \mathbb{C}^{2^{N/2}}$. Algebraically, they generate a complex Clifford algebra.

The complex Clifford algebra is the unital associative algebra generated by symbols ψ_i subject to (1.4); the adjoint makes these generators self-adjoint. Irreducible means that no nonzero proper subspace is invariant under every generator. The Fock representation has this property, as proved in Appendix A.6. A qubit is a factor $\mathbb{C}^2$; equality of Hilbert-space dimensions does not identify its local observables with fermionic ones.

A product of q distinct Majoranas has a simple adjoint rule. If $X = \psi_{i_1} \cdots \psi_{i_q}$, $i_1 < \cdots < i_q$, then $X^* = (-1)^{q(q-1)/2} X$. For $q = 4$, the sign is $+1$, so $\psi_i\psi_j\psi_k\psi_l$ is self-adjoint. For every even q, the phase in $\psi_I = i^{q/2}\psi_{i_1} \cdots \psi_{i_q}$ makes ψ_I self-adjoint; moreover $\psi_I^2 = I$ in the normalization used here.

Jordan-Wigner representation. For numerical intuition, one may represent Majoranas using Pauli matrices.

Our matrices are

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The subscript a means that the indicated matrix acts on tensor factor a and the identity acts on the other factors. Products with no factors equal the identity. This choice of σ^y identifies the Fock vacuum with the all-down spin state for our convention $c_a = (\psi_{2a-1} - i\psi_{2a})/2$. On $(\mathbb{C}^2)^{\otimes M}$,

$$\psi_{2a-1} = \left(\prod_{b<a} \sigma_b^z\right) \sigma_a^x,$$
$$\psi_{2a} = \left(\prod_{b<a} \sigma_b^z\right) \sigma_a^y.$$

The strings of σ^z are precisely what generate anticommutation between different modes. This realizes the Clifford generators as concrete matrices on a tensor-product Hilbert space.

Free fermions and quadratic Hamiltonians. A general quadratic Majorana Hamiltonian can be written as

$$H_2 = \frac{i}{4} \sum_{i,j=1}^{N} A_{ij}\psi_i\psi_j, \qquad A^T = -A, \qquad A \in M_N(\mathbb{R}).$$

This includes the $q = 2$ SYK model: in that case the random couplings are simply the entries of the real skew-symmetric matrix A. Its diagonalization follows from the canonical form of a real skew-symmetric matrix. For $N = 2M$, there are $O \in O(N)$ and numbers $\varepsilon_a \geq 0$ such that

$$OAO^T = \bigoplus_{a=1}^{M} \begin{pmatrix} 0 & -\varepsilon_a \\ \varepsilon_a & 0 \end{pmatrix}. \tag{1.5}$$

Thus the one-particle eigenvalues of A are the pairs $\pm i\varepsilon_a$. If

$$\chi_i = \sum_j O_{ij}\psi_j,$$

then orthogonality gives $\{\chi_i, \chi_j\} = 2\delta_{ij}I$, so this change of basis preserves the Majorana CAR. Pairing the transformed Majoranas into complex fermions,

$$d_a = \frac{\chi_{2a-1} - i\chi_{2a}}{2}, \qquad n_a = d_a^\dagger d_a,$$

gives $i\chi_{2a-1}\chi_{2a} = 1 - 2n_a$ and hence

$$H_2 = \sum_{a=1}^{M} \varepsilon_a \left(n_a - \frac{1}{2} \right). \tag{1.6}$$

In particular, $[n_a, n_b] = 0$. There is a simultaneous eigenbasis of occupation-number states $|n_1, \ldots, n_M\rangle$, $n_a \in \{0, 1\}$, with the additive energy

$$E(n_1, \ldots, n_M) = \sum_{a=1}^{M} \varepsilon_a \left(n_a - \frac{1}{2} \right).$$

Therefore diagonalizing the exponentially large $2^M \times 2^M$ many-body Hamiltonian reduces to diagonalizing the $N \times N$ matrix A. For example, the partition function factorizes as

$$Z_2(\beta) = \prod_{a=1}^{M} 2\cosh\left(\frac{\beta\varepsilon_a}{2} \right),$$

and all thermal correlation functions follow from the two-point function by Wick's theorem. This is the precise sense in which $q = 2$ is a free-fermion, and hence explicitly solvable, model.

Here free means quadratic in the fermionic generators. Wick's theorem states that an even correlation of linear fermionic operators is the signed sum of products of two-point correlations over all pairings, and odd correlations vanish. A state with this property is called Gaussian or quasifree. A proof for this quadratic Gibbs state and the commutator calculation below are given in Appendix A.6. For the $q = 2$ Hamiltonian in (1.1), the entries are $A_{ij} = 2\sigma_N g_{ij}$ for $i < j$.

The same simplicity is visible dynamically. The commutator of a quadratic Hamiltonian with a Majorana operator is again linear:

$$[H_2, \psi_k] = -i \sum_j A_{kj}\psi_j.$$

Consequently the Heisenberg equations close on the N-dimensional one-particle space,

$$\frac{d}{dt}\psi(t) = A\psi(t), \qquad \psi(t) = e^{tA}\psi(0).$$

No new many-body operators are generated by time evolution.

For interacting SYK, the same argument no longer applies. If $q \geq 4$, then $[H_q, \psi_m]$ is generically a linear combination of products of $q-1$ Majoranas, rather than a linear combination of the ψ_j. An orthogonal change of one-particle basis preserves polynomial degree: it can put a two-form $A \in \Lambda^2\mathbb{R}^N$ into the block form (1.5), but it cannot turn a generic tensor in $\Lambda^q\mathbb{R}^N$ into a quadratic Hamiltonian. The Heisenberg evolution therefore mixes increasingly complicated operator sectors, and the reduction to independent occupation numbers fails. This is why the strongly interacting low-energy state need not admit a quasiparticle interpretation.

1.3. Grassmann variables and thermal correlators

The operator formulation is fundamental. Grassmann variables are a powerful algebraic device for reorganizing traces and correlation functions.

Grassmann algebra and Berezin integration. A Grassmann variable is a formal exterior-algebra generator, not a random scalar and not a Clifford matrix. In particular its square is zero, whereas a Majorana matrix here squares to I. With the ordered differential $d\theta_n \cdots d\theta_1$, the integral extracts the coefficient of $\theta_1 \cdots \theta_n$.

Let $\theta_1, \ldots, \theta_n$ generate a Grassmann algebra: $\theta_i\theta_j = -\theta_j\theta_i$. In particular, $\theta_i^2 = 0$. Berezin integration is defined by

$$\int d\theta\, 1 = 0, \qquad \int d\theta\, \theta = 1.$$

It is differentiation and integration at the same time because every polynomial is affine in each Grassmann generator.

For an antisymmetric $2m \times 2m$ matrix A,

$$\int d\theta_{2m} \cdots d\theta_1 \exp\left(\frac{1}{2}\theta^T A \theta\right) = \mathrm{Pf}(A), \tag{1.7}$$

with $\mathrm{Pf}(A)^2 = \det A$.

The Pfaffian is the polynomial

$$\mathrm{Pf}(A) = \frac{1}{2^m m!} \sum_{\pi \in S_{2m}} \mathrm{sgn}(\pi) \prod_{a=1}^{m} A_{\pi(2a-1),\pi(2a)}.$$

The sum is over permutations of $2m$ labels. Expanding the exponential and extracting its top coefficient proves (1.7); the determinant identity is proved in Appendix A.6. For comparison, if A is a real symmetric positive-definite $n \times n$ matrix, the ordinary Gaussian integral is

$$\int_{\mathbb{R}^n} e^{-x^T A x/2}\, dx = (2\pi)^{n/2} (\det A)^{-1/2}.$$

The determinant occurs with the opposite power for fermions. This algebraic fact is the source of the logarithmic determinant in the large-N collective action.

Finite-temperature quantum mechanics. The partition function Z normalizes the Gibbs state ρ. The total free energy, internal energy, entropy, and heat capacity at fixed Hamiltonian are

$$F = -\beta^{-1} \log Z, \quad U = \operatorname{Tr}(\rho H), \quad S = -\operatorname{Tr}(\rho \log \rho), \quad C_V = \frac{dU}{dT}.$$

The free-energy density is F/N, and pressure here means $N^{-1} \log Z$, without a spatial volume. A correlator is the expectation of a product of operators. Imaginary or Euclidean time is the real variable τ obtained formally by $t = -i\tau$; $e^{\tau H}$ need not be unitary.

For a fixed disorder realization $g = (g_I)_{I \in \mathcal{I}_N}$, define

$$Z_g(\beta) = \operatorname{Tr} e^{-\beta H_q}, \qquad \rho_{\beta,g} = \frac{e^{-\beta H_q}}{Z_g(\beta)}.$$

For an observable A, $\langle A \rangle_{\beta,g} = \operatorname{Tr}(\rho_{\beta,g} A)$. Imaginary-time evolution is $A(\tau) = e^{\tau H} A e^{-\tau H}$, $0 \le \tau < \beta$. The Euclidean two-point function is

$$G_{ij,g}(\tau) = \langle T_\tau \psi_i(\tau) \psi_j(0) \rangle_{\beta,g}.$$

Fermionic time ordering moves the later operator to the left with a minus sign for each exchange of two odd operators:

$$T_\tau A(\tau) B(0) = \begin{cases} A(\tau)B(0), & \tau > 0, \\ -B(0)A(\tau), & \tau < 0. \end{cases}$$

Equal-time values mean one-sided limits. In particular $G_{ii}(0^+) = 1$, $G_{ii}(0^-) = -1$. A Green function or propagator in these notes is this two-point kernel, with the specified sign convention.

After disorder averaging and flavor averaging,

$$G_N(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}_g G_{ii,g}(\tau). \tag{1.8}$$

Fermionic thermal correlators are antiperiodic, $G_N(\tau + \beta) = -G_N(\tau)$, which implies the Matsubara frequencies

$$\omega_n = \frac{(2n+1)\pi}{\beta}.$$

Thus

$$G(\tau) = \frac{1}{\beta} \sum_{n \in \mathbb{Z}} e^{-i\omega_n \tau} G(i\omega_n).$$

Spectral representation. A spectral measure records energy differences and the matrix elements that couple them. The notation δ_a is the unit point mass at a; it differs from a density with respect to Lebesgue measure. The Lehmann representation means insertion of a complete energy eigenbasis. Retarded means that the response vanishes for $t < 0$; $\theta(t)$ is the Heaviside function. Its Fourier–Laplace transform uses e^{izt} for $\operatorname{Im} z > 0$. The thermal and spectral identities below are proved in Appendix A.7.

Fix the disorder and a self-adjoint Majorana operator ψ. Let $H|n\rangle = E_n|n\rangle$ and $Z = \sum_n e^{-\beta E_n}$. Its positive spectral measure is

$$d\nu_\psi(\omega) = \frac{1}{Z}\sum_{m,n}(e^{-\beta E_n} + e^{-\beta E_m})|\langle n|\psi|m\rangle|^2\,\delta_{E_m-E_n}(d\omega). \tag{1.9}$$

In particular, $\nu_\psi(\mathbb{R}) = \langle\{\psi,\psi\}\rangle_\beta = 2$. With our convention $G(\tau) = \langle T_\tau\psi(\tau)\psi(0)\rangle_\beta$ and Fourier transform $G(i\omega_n) = \int_0^\beta e^{i\omega_n\tau}G(\tau)\,d\tau$, the spectral formulas are

$$G(i\omega_n) = -\int_{\mathbb{R}}\frac{d\nu_\psi(\omega)}{i\omega_n-\omega}, \qquad G^R(z) = \int_{\mathbb{R}}\frac{d\nu_\psi(\omega)}{z-\omega}, \quad \operatorname{Im} z > 0. \tag{1.10}$$

Here $G^R(t) = -i\theta(t)\langle\{\psi(t),\psi(0)\}\rangle_\beta$. Thus the retarded function is the continuation of $-G$, rather than G, in this Euclidean sign convention. If the spectral measure has a density A, then $A(\omega) = -\pi^{-1}\operatorname{Im} G^R(\omega + i0^+)$. At finite N, this identity is understood distributionally because the measure is atomic. Appendix A gives the corresponding imaginary-time expansion.

1.4. The SYK Hamiltonian

Definition and normalization. Fix even integers N and q with $2 \le q \le N$, and let $\mathcal{I}_N = \{(i_1,\dots,i_q) : 1 \le i_1 < \cdots < i_q \le N\}$. For $I = (i_1,\dots,i_q) \in \mathcal{I}_N$, set $\psi_I := i^{q/2}\psi_{i_1}\cdots\psi_{i_q}$. Let $(g_I)_{I\in\mathcal{I}_N}$ be iid standard normal random variables and define

$$H_q = \sigma_N\sum_{I\in\mathcal{I}_N} g_I\psi_I, \qquad \sigma_N^2 = \frac{(q-1)!}{N^{q-1}}. \tag{1.11}$$

Equivalently, the random coefficients $J_I := \sigma_N g_I$ are independent centered Gaussians with

$$\mathbb{E}[J_I J_{I'}] = \frac{(q-1)!}{N^{q-1}}\delta_{I,I'}. \tag{1.12}$$

Because q is even and $\{\psi_i,\psi_j\} = 2\delta_{ij}I$, each ψ_I is a self-adjoint unitary. For $q = 4$, formula (1.11) differs by an irrelevant overall sign from writing the interaction as a sum of $\psi_i\psi_j\psi_k\psi_l$, since the centered Gaussian law is invariant under $g_I \mapsto -g_I$.

Example 1.3 (Two Majorana operators). For $N = q = 2$, take $\psi_1 = \sigma^x$ and $\psi_2 = \sigma^y$. Since $i\psi_1\psi_2 = -\sigma^z$, the Hamiltonian and partition function are

$$H_2 = -\frac{g}{\sqrt{2}}\sigma^z, \qquad Z_g(\beta) = 2\cosh\left(\frac{\beta g}{\sqrt{2}}\right), \qquad g \sim \mathcal{N}(0,1).$$

Thus thermal expectation is a trace over two states, while disorder expectation is an integral against one Gaussian variable. These are the two averages that remain distinct in the general model.

Why the variance scales as $N^{-(q-1)}$. There are

$$\binom{N}{q} \sim \frac{N^q}{q!}$$

interaction monomials. The variance scaling is chosen so that the large-N self-energy remains of order one and the total free energy is of order N. In diagrammatic language, each disorder contraction contributes $N^{-(q-1)}$, while free flavor-index sums contribute powers of N. The competition produces a nontrivial $N \to \infty$ limit rather than either a free theory or a divergent one.

A second-moment computation makes the scale precise. Since $\psi_I^2 = I$ and distinct Clifford monomials are orthogonal under the normalized trace,

$$2^{-N/2}\mathbb{E}\,\mathrm{Tr}\,H_q^2 = \sigma_N^2\binom{N}{q} = \frac{N}{q}\prod_{r=1}^{q-1}\Big(1-\frac{r}{N}\Big) \sim \frac{N}{q}.$$

The root-mean-square eigenvalue, averaged over disorder and normalized trace, is therefore of order $\sqrt{N}$. This calculation does not determine the spectral edge. The order-N ground-state scale requires a separate argument; see Section 5.

Quenched and annealed thermodynamics. The quenched free-energy density is

$$f_N^{\mathrm{q}}(\beta) = -\frac{1}{\beta N}\mathbb{E}\log Z_g(\beta).$$

The annealed free-energy density is

$$f_N^{\mathrm{a}}(\beta) = -\frac{1}{\beta N}\log \mathbb{E}Z_g(\beta).$$

By Jensen, $\mathbb{E}\log Z \le \log \mathbb{E}Z$, so $f_N^{\mathrm{a}}(\beta) \le f_N^{\mathrm{q}}(\beta)$. Concentration of the sample pressure is elementary in the Gaussian model. It does not, by itself, imply equality of the quenched and annealed limits.

Proposition 1.4 (Concentration of the pressure). *For $\beta > 0$, put $P_N(g) = N^{-1}\log Z_g(\beta)$. Then*

$$\mathrm{Var}(P_N) \le \frac{\beta^2}{qN}, \qquad \mathbb{P}\big(|P_N - \mathbb{E}P_N| \ge t\big) \le 2\exp\left(-\frac{qNt^2}{2\beta^2}\right), \quad t > 0.$$

In particular, $P_N - \mathbb{E}P_N \to 0$ in probability at every fixed β.

Proof. Differentiation of the matrix exponential under the trace, followed by cyclicity, gives

$$\partial_{g_I} P_N = -\frac{\beta\sigma_N}{N}\langle\psi_I\rangle_{\beta,g}.$$

Since $\|\psi_I\| = 1$,

$$\|\nabla P_N\|_2^2 \le \frac{\beta^2\sigma_N^2}{N^2}\binom{N}{q} = \frac{\beta^2}{qN}\prod_{r=1}^{q-1}\Big(1-\frac{r}{N}\Big) \le \frac{\beta^2}{qN}.$$

The Gaussian Poincaré inequality proves the variance bound, and Gaussian concentration for Lipschitz functions proves the tail bound. □

Remark 1.5 (Dependence on temperature). The pressure bound in Proposition 1.4 gives concentration along every sequence $\beta_N = o(\sqrt{N})$. This is a sufficient range for that bound, not a physical crossover.

Indeed, the sample free-energy density $f_{N,g}(\beta) = -P_N(g)/\beta$ satisfies

$$\mathrm{Var}(f_{N,g}(\beta)) \le \frac{1}{qN}, \qquad \mathbb{P}\big(|f_{N,g}(\beta) - \mathbb{E}f_{N,g}(\beta)| \ge t\big) \le 2e^{-qNt^2/2}$$

for every $\beta > 0$. Thus its concentration persists for arbitrary deterministic temperature sequences. The loss of a useful estimate for one observable does not imply a loss of self-averaging for all observables, nor does it locate the Schwarzian crossover.

Macroscopic self-averaging is compatible with microscopic random-matrix fluctuations. The free energy can concentrate while individual level spacings remain random.

2. The large-N equations and the infrared limit

2.1. The melonic large-N limit

The key solvability mechanism is a combinatorial selection rule on Feynman graphs.

Disorder averaging. A Feynman graph encodes a term in a formal expansion of a correlator: a vertex is an interaction, an edge a two-point contraction, and an internal label is summed. A disorder contraction is instead a Gaussian pairing of two coupling variables. Wick's Gaussian formula follows by differentiating $\mathbb{E}e^{\sum_I t_I g_I} = e^{\sum_I t_I^2/2}$: odd moments vanish and each even moment is a sum over pairings of Kronecker deltas. If an averaged graph has v disorder pairings and ℓ independent flavor sums, its naive power is $N^{\ell-(q-1)v}$, before external normalization. Restrictions to distinct labels require separate counting.

In a perturbative expansion, each interaction vertex carries a random coefficient $\sigma_N g_I$. Since the disorder is Gaussian, averaging pairs the couplings: $\mathbb{E}[(\sigma_N g_I)(\sigma_N g_{I'})] = \sigma_N^2 \delta_{I,I'}$. After averaging, every pair of interaction vertices comes with identified flavor indices. The resulting graph carries a power of N determined by how many independent flavor-index loops remain.

The leading graphs are the melonic graphs. They are recursively generated by replacing a propagator line with a two-vertex self-energy insertion and iterating.

Formal statement 2.1 (Melonic dominance). At fixed q and fixed perturbative order, the leading power of N is attained by melonic graphs. Nonmelonic graphs are suppressed by powers of $1/N$.

For SYK, melonic dominance has a rigorous combinatorial diagrammatic proof [15]. What remains formal is the passage from order-by-order diagrammatics to the full nonperturbative infrared theory in the strongest sense used in physics.

Details omitted: the general graph-counting theorem and any uniform summation of its expansion. A melon has two interaction vertices joined by $q-1$ fermion lines and one disorder pairing, with two remaining external fermion ends; replacing an edge by this insertion defines the recursive family.

Dyson equation. The self-energy is defined, when the inverses exist, by $\Sigma = G_0^{-1} - G^{-1}$. Thus the Dyson identity by itself is an algebraic definition, whereas $\Sigma = G^{q-1}$ is the substantive large-N closure. Diagrammatically a self-energy insertion is a two-ended contribution not separable by cutting one internal propagator. Schwinger–Dyson equations are identities between correlators obtained from equations of motion or integration by parts; closing the hierarchy requires an additional argument.

Under $\{\psi_i, \psi_j\} = 2\delta_{ij} I$, one has $G_0(0^+) = 1$ and the free Majorana propagator is $G_0(i\omega_n) = 2(-i\omega_n)^{-1}$. The exact Dyson relation is $G(i\omega_n)^{-1} = G_0(i\omega_n)^{-1} - \Sigma(i\omega_n)$. With the convention used here,

$$G(i\omega_n)^{-1} = -\frac{i\omega_n}{2} - \Sigma(i\omega_n). \tag{2.1}$$

Melonic closure gives for quartic SYK

$$\Sigma(\tau) = G(\tau)^3. \tag{2.2}$$

For general even q, $\Sigma(\tau) = G(\tau)^{q-1}$. The unit coefficient follows from using both $\sigma_N^2 = (q-1)!/N^{q-1}$ and Clifford generators squaring to I.

Equations (2.1) and (2.2) are the central large-N Schwinger-Dyson equations.

Kernel form. Regard G and Σ as integral kernels. Then $(\frac{1}{2}\partial_\tau - \Sigma) \circ G = I$, meaning

$$\int_0^\beta d\tau' \left[\frac{1}{2}\partial_{\tau_1}\delta(\tau_1 - \tau') - \Sigma(\tau_1, \tau')\right] G(\tau', \tau_2) = \delta(\tau_1 - \tau_2).$$

At a translation-invariant thermal saddle, $G(\tau_1, \tau_2) = G(\tau_1 - \tau_2)$. Thus a many-body problem on a Hilbert space of dimension $2^{N/2}$ becomes a nonlinear integral equation for one scalar function of time.

2.2. The bilocal collective action

There is a second derivation of the same equations which is closer to mean-field variational methods.

Introducing G and Σ. Bilocal means depending on two independent time arguments. Here both kernels are antisymmetric, $G_{12} = -G_{21}$ and $\Sigma_{12} = -\Sigma_{21}$, where 12 abbreviates (τ_1, τ_2). Kernel composition is $(A \circ B)_{12} = \int_0^\beta A_{13} B_{32}\, d\tau_3$. The delta kernel and derivative act on antiperiodic functions on the thermal circle. The symbols $\mathcal{D}G\,\mathcal{D}\Sigma$ denote formal functional integration, not a specified Lebesgue measure on a function space.

Formally insert the constraint

$$G(\tau_1, \tau_2) = \frac{1}{N}\sum_{i=1}^{N} \psi_i(\tau_1)\psi_i(\tau_2)$$

using a bilocal Lagrange multiplier $\Sigma(\tau_1, \tau_2)$. After Gaussian disorder averaging and integrating out the fermions, one obtains for general even q

$$\frac{I_{\text{eff}}[G, \Sigma]}{N} = -\frac{1}{2}\log\det\left(\tfrac{1}{2}\partial_\tau - \Sigma\right) + \frac{1}{2}\int d\tau_1 d\tau_2 \left[\Sigma G - \frac{1}{q}G^q\right]. \tag{2.3}$$

The determinant term is the logarithm of a Pfaffian in Majorana language. Equation (1.7) is the finite-dimensional algebraic prototype.

For a single copy, the disorder average gives the annealed partition function in the formal form

$$\mathbb{E}Z \simeq \int \mathcal{D}G\,\mathcal{D}\Sigma\, \exp\left[-N\mathcal{I}(G, \Sigma)\right].$$

Here $\mathcal{I} = I_{\text{eff}}/N$, and a regularization and integration contour are implicit. Quenched observables require replicas or a separate justification that the relevant saddle gives their limit. See [3, 5] for the collective-field calculation.

Stationary equations. Varying (2.3) with respect to G gives $\Sigma(\tau_1, \tau_2) = G(\tau_1, \tau_2)^{q-1}$. Varying with respect to Σ gives $G = (\frac{1}{2}\partial_\tau - \Sigma)^{-1}$. Hence the saddle equations are precisely the Schwinger-Dyson equations.

Remark 2.2 (Analogy with a Laplace principle). The schematic similarity

$$\int e^{-NI(x)} dx \sim e^{-N \inf I}$$

is useful, but the fermionic functional integral is not automatically a rigorous probability measure. A mathematical derivation requires control of disorder averaging, the collective-variable transformation, saddle selection, and low-temperature limits.

Large-N factorization. A singlet is invariant under the flavor group $O(N)$; $N^{-1}\sum_i \psi_i(\tau_1)\psi_i(\tau_2)$ is an example. Connected means subtracting products associated with disconnected partitions. For two composite observables it is their covariance. For four centered odd insertions, with $G_{ab} = \langle T_\tau A_a A_b\rangle$, it is

$$\langle T_\tau A_1 A_2 A_3 A_4\rangle_c = \langle T_\tau A_1 A_2 A_3 A_4\rangle - G_{12}G_{34} + G_{13}G_{24} - G_{14}G_{23}.$$

One must specify whether this subtraction precedes disorder averaging; the two operations can differ by disorder covariances. The $1/N$ saddle argument assumes an invertible Hessian after removing redundancies; Appendix A.9 explains the scaling. Details omitted: a uniform microscopic factorization estimate.

One consequence of the prefactor N is that fluctuations of suitably normalized singlet observables are suppressed. Schematically,

$$\langle \mathcal{O}_1 \mathcal{O}_2\rangle - \langle \mathcal{O}_1\rangle\langle \mathcal{O}_2\rangle = O(N^{-1}).$$

This is the analogue of concentration around a mean-field saddle. The connected four-point function is therefore naturally an $O(1/N)$ object.

2.3. The infrared conformal limit

We now take $q \geq 4$ and examine the formal low-frequency approximation to the Schwinger–Dyson equations. The approximation is made after the large-N reduction [3].

Dropping the kinetic term. Infrared (IR) means low frequency or long time compared with the interaction scale: $|\omega| \ll J$ or $|\tau| \gg J^{-1}$. Ultraviolet (UV) means the opposite range. Strong coupling at temperature T means $\beta J \gg 1$. A scaling dimension Δ specifies the transformation factor of a field under time dilation; its two-point function scales as $|\tau|^{-2\Delta}$. An anomalous dimension differs from the free-theory value, which is zero for this free Majorana correlator away from coincident times. A fixed point is a description unchanged by the appropriate rescaling of time and fields.

The full equation is

$$G(i\omega)^{-1} = -\frac{i\omega}{2} - \Sigma(i\omega).$$

If $|\omega| \ll 1$, and if the self-energy dominates $i\omega$, one may first solve the scale-invariant equation obtained by dropping the kinetic term. In time-domain kernel notation,

$$\Sigma \circ G = -I. \tag{2.4}$$

Together with $\Sigma(\tau) = G(\tau)^{q-1}$, this admits a power-law solution.

Assume

$$G_c(\tau) = b\,\frac{\operatorname{sgn}\tau}{|\tau|^{2\Delta}}.$$

Then

$$\Sigma_c(\tau) \propto \frac{\operatorname{sgn}\tau}{|\tau|^{2\Delta(q-1)}}.$$

The convolution in (2.4) must scale as a delta distribution. Since $\delta(\lambda\tau) = \lambda^{-1}\delta(\tau)$, dimensional consistency gives $2\Delta(q-1) + 2\Delta = 2$, so

$$\Delta = \frac{1}{q}.$$

For $q = 4$,

$$\Delta = \frac{1}{4}, \qquad G_c(\tau) = b\frac{\operatorname{sgn}\tau}{|\tau|^{1/2}}. \tag{2.5}$$

With the normalization used here,

$$b^4 = \frac{1}{4\pi}.$$

More generally $b^q = (1-2/q)\tan(\pi/q)/(2\pi)$ with $b > 0$. This follows by Fourier transforming homogeneous distributions, not by an absolutely convergent convolution integral. Appendix A.8 supplies the regularization and coefficient calculation. It also verifies the finite-temperature transformation and reparametrization covariance. These computations solve the formal IR equations; details omitted: error bounds comparing them with microscopic SYK.

Finite temperature. The conformal map from the line to the thermal circle suggests

$$G_{\beta,c}(\tau) = b\,\operatorname{sgn}(\tau)\left[\frac{\pi}{\beta\sin(\pi|\tau|/\beta)}\right]^{2\Delta}, \qquad 0 < |\tau| < \beta. \tag{2.6}$$

For $q = 4$, the exponent is $1/2$.

The physical interpretation is that strong interactions generate an anomalous scaling dimension. The fermion field does not behave like a weakly dressed free particle.

Emergent reparametrization covariance. An orientation-preserving reparametrization is a smooth change of coordinate with $f' > 0$. On the circle we use a lift $f : \mathbb{R} \to \mathbb{R}$ with $f(\tau+\beta) = f(\tau) + \beta$, modulo shifts by β. The resulting group is $\mathrm{Diff}^+(S^1)$, abbreviated $\mathrm{Diff}(S^1)$ here. Also $PSL(2,\mathbb{R}) = SL(2,\mathbb{R})/\{I,-I\}$ acts by fractional linear transformations on the real projective line. On the thermal circle this action is on $\tan(\pi f/\beta)$, not directly on the lift f. Covariance means solutions transform into solutions, whereas invariance of one saddle means its kernel is unchanged. The quotient identifies configurations related by this projective action.

The infrared equations are covariant under orientation-preserving time reparametrizations $\tau \mapsto f(\tau)$ if

$$G(\tau_1, \tau_2) \mapsto [f'(\tau_1) f'(\tau_2)]^{\Delta} G(f(\tau_1), f(\tau_2)), \tag{2.7}$$

while

$$\Sigma(\tau_1, \tau_2) \mapsto [f'(\tau_1) f'(\tau_2)]^{1-\Delta} \Sigma(f(\tau_1), f(\tau_2)).$$

A particular conformal saddle is invariant only under the Möbius subgroup

$$f(\tau) = \frac{a\tau + b}{c\tau + d}, \qquad ad - bc = 1.$$

Thus the symmetry pattern is often written

$$\mathrm{Diff}(S^1) \longrightarrow PSL(2, \mathbb{R}).$$

The full microscopic theory is not exactly reparametrization invariant. The kinetic term explicitly breaks the symmetry. Therefore the reparametrization modes are soft rather than exact zero modes.

2.4. The Schwarzian soft mode

The Schwarzian derivative. For a C^3 function f with f' nowhere zero define

$$\{f, \tau\} = \frac{f'''(\tau)}{f'(\tau)} - \frac{3}{2} \left(\frac{f''(\tau)}{f'(\tau)} \right)^2. \tag{2.8}$$

It satisfies the projective invariance

$$\left\{ \frac{af + b}{cf + d}, \tau \right\} = \{f, \tau\}.$$

Hence $PSL(2, \mathbb{R})$ is naturally quotiented out.

The leading low-energy action for reparametrization fluctuations has the form

$$I_{\mathrm{Sch}}[f] = -N\alpha_S \int_0^\beta d\tau \left\{ \tan\left(\frac{\pi f(\tau)}{\beta} \right), \tau \right\}, \tag{2.9}$$

Here $\alpha_S > 0$ is a model-dependent constant, and we write $C = N\alpha_S$ in the units of these notes. For orientation-preserving maps with $f(\tau + \beta) = f(\tau) + \beta$, the formal configuration space is $\mathrm{Diff}(S^1)/PSL(2, \mathbb{R})$.

Why the Schwarzian matters. A zero mode is a vector in the kernel of the second variation of an action. A soft mode has a small restoring coefficient in the scaling under discussion; a massive direction has a nonzero restoring coefficient at that scale. These terms describe collective fluctuations and do not assert the existence of a particle mass in the finite SYK Hilbert space. Details omitted: integrating the other bilocal directions and matching α_S to the microscopic theory.

The strict conformal approximation produces a nearly flat manifold of reparametrized saddles. The small UV kinetic term lifts this degeneracy. The most infrared-singular fluctuations therefore come from motion along this almost-flat manifold, not from generic massive fluctuations. Integrating out the massive directions produces the Schwarzian action.

The dimensionless fluctuation parameter. Put $u = \tau/\beta$ and $f(\tau) = \beta g(u)$, where $g(u+1) = g(u)+1$. The chain rule for the Schwarzian gives

$$I_{\mathrm{Sch}}[f] = \frac{C}{\beta}\,\mathcal{A}[g], \qquad \mathcal{A}[g] \coloneqq -\int_0^1 \{\tan(\pi g(u)), u\}\, du. \tag{2.10}$$

This identifies β/C as the fluctuation parameter. For a fixed smooth periodic function h, set $g(u) = u + \varepsilon h(u)$. Integration by parts gives

$$\mathcal{A}[g] = \frac{1}{2}\int_0^1 \left(\frac{g''}{g'}\right)^2 du - 2\pi^2 \int_0^1 (g')^2 du,$$
$$\mathcal{A}[u+\varepsilon h] = -2\pi^2 + \frac{\varepsilon^2}{2}\int_0^1 \left((h'')^2 - 4\pi^2 (h')^2\right) du + O_h(\varepsilon^3).$$

The quadratic form vanishes on the Fourier modes with indices $0, 1, -1$, which generate the projective redundancy.

Explicitly, for $h(u) = \sum_n h_n e^{2\pi i n u}$ real and smooth, Parseval's identity gives

$$\frac{1}{2}\int_0^1 \left((h'')^2 - 4\pi^2 (h')^2\right) du = 8\pi^4 \sum_{n\in\mathbb{Z}} n^2(n^2-1)|h_n|^2.$$

The three real modes are 1, $\cos(2\pi u)$, and $\sin(2\pi u)$. On the remaining modes it is positive. The Gaussian approximation therefore assigns dimensionless fluctuations of size $(\beta/C)^{1/2}$ to a fixed nonzero mode after this redundancy is removed.

For $\beta/C \to 0$, expansion around one thermal saddle is appropriate. When β/C is of order one, the soft coordinate must be integrated over. This is a singular saddle-point problem: the leading collective action has a family of stationary points, and its small explicit symmetry breaking determines an action along that family. The coordinate f parametrizes collective configurations; it is not an additional microscopic fermion.

Low-temperature thermodynamics. At leading order in N, in the semiclassical range $1 \ll \beta \ll N$, the Schwarzian theory gives $F(T) = E_0 - S_0 T - cT^2 + \cdots$, so

$$S(T) = -\frac{\partial F}{\partial T} = S_0 + 2cT + \cdots .$$

For SYK at large N,

$$S(T) = N s_0 + N\gamma T + \cdots . \tag{2.11}$$

The residual entropy density s_0 is nonzero in the order of limits

$$\lim_{T\downarrow 0}\lim_{N\to\infty} \frac{S_N(T)}{N} = s_0 > 0.$$

If the ground-state multiplicity $d_{0,N}$ satisfies $\log d_{0,N} = o(N)$, as expected for generic nonsupersymmetric SYK after symmetry degeneracies, then

$$\lim_{N\to\infty}\lim_{T\downarrow 0} \frac{S_N(T)}{N} = 0$$

in the generic situation, up to symmetry degeneracies. The two limits do not commute.

The finite-dimensional identity $S_N(T) \to \log d_{0,N}$ is proved below. Details omitted: a generic multiplicity theorem and the microscopic derivation of the positive residual entropy density.

The quantum Schwarzian partition function. The Schwarzian functional integral can be evaluated beyond the small-fluctuation regime [20]. Choose its overall normalization so that

$$Z_{\mathrm{Sch}}(\beta; C) = \frac{1}{\sqrt{2\pi}} \left(\frac{C}{\beta}\right)^{3/2} e^{2\pi^2 C/\beta}. \tag{2.12}$$

Its Laplace density is then

$$\rho_{\mathrm{Sch}}(\varepsilon; C) = \frac{C}{2\pi^2} \sinh\bigl(2\pi\sqrt{2C\varepsilon}\bigr), \qquad \varepsilon \geq 0. \tag{2.13}$$

Indeed, for $a, \beta > 0$,

$$\int_0^\infty e^{-\beta\varepsilon} \sinh\bigl(a\sqrt{\varepsilon}\bigr)\, d\varepsilon = \frac{a\sqrt{\pi}}{2\beta^{3/2}} e^{a^2/(4\beta)},$$

as follows by differentiating the Gaussian integral after setting $\varepsilon = x^2$. Taking $a = 2\pi\sqrt{2C}$ verifies the normalization.

Let E_0 and $S_0 = N s_0(q) + o(N)$ be the smooth edge and entropy parameters. Subextensive terms in S_0 depend on normalization. The effective partition function is

$$Z_{\mathrm{smooth}}(\beta) = e^{S_0 - \beta E_0} Z_{\mathrm{Sch}}(\beta; C), \qquad \rho_{\mathrm{smooth}}(E_0 + \varepsilon) = e^{S_0} \rho_{\mathrm{Sch}}(\varepsilon; C). \tag{2.14}$$

Here S_0 is already extensive. A different overall normalization of Z_{Sch} must be accompanied by the corresponding change in S_0. In particular,

$$\begin{aligned} \log Z_{\mathrm{smooth}} &= S_0 - \beta E_0 + \frac{2\pi^2 C}{\beta} + \frac{3}{2} \log \frac{C}{\beta} - \frac{1}{2} \log(2\pi), \\ -\partial_\beta \log Z_{\mathrm{smooth}} &= E_0 + \frac{2\pi^2 C}{\beta^2} + \frac{3}{2\beta}. \end{aligned}$$

The term $\frac{3}{2} \log C$ cannot be absorbed into a constant independent of N. At leading semiclassical order these formulas give $c = 2\pi^2 C$ and $\gamma = 4\pi^2 \alpha_S$ in (2.11).

Formula (2.12) is exact within the Schwarzian theory, with its specified measure.

Details omitted: the definition of that functional measure and the localization proof of its evaluation. The elementary Laplace transform and thermodynamic consequences are verified in Appendix A.7. It is an approximation to microscopic SYK only in a regime where matching to that theory is justified. Its smooth Laplace density also differs from the atomic spectral measure of a finite matrix. For $\beta \gg C$, the Schwarzian coordinate fluctuates strongly, while a smooth spectral description can still be useful if the relevant energy window contains many levels.

A hierarchy of large-N and low-temperature regimes. The phrases "large N" and "low temperature" do not specify a unique asymptotic regime. Their meaning depends on how the inverse temperature β grows with N, and the corresponding changes should not all be called phase transitions. To restore an interaction scale, replace H_q by JH_q with $J > 0$. Then $C = N\alpha_S/J$, and the dimensionless inverse temperature is βJ. We otherwise retain $J = 1$ and the convention $\psi_i^2 = I$. The principal ranges, with the last crossover estimated from the smooth density, are

inverse-temperature scale	effective description
$\beta J = O(1)$	ordinary large-N Schwinger–Dyson theory
$1 \ll \beta J \ll N$	conformal SYK and a semiclassical Schwarzian mode
$\beta J \asymp N$	Schwarzian fluctuations become order one
$N \ll \beta J \ll N e^{2S_0/3}$	quantum Schwarzian with an effectively continuous spectrum
$\beta J \asymp N e^{2S_0/3}$	the microscopic spectral edge becomes visible
$\beta J \gg N e^{2S_0/3}$	the lowest individual eigenvalues dominate

Here S_0 is as in (2.14). For the standard quartic Majorana model, the large-N saddle calculation gives $s_0(4) \simeq 0.2324$ [3]; details omitted: its numerical evaluation. The scales $\beta J \asymp 1$ and $\beta J \asymp N$ describe crossovers between asymptotic approximations, not thermodynamic phase transitions. The first is the UV-to-IR crossover. At the second, the Schwarzian coupling $C \asymp N/J$ satisfies $\beta/C = O(1)$, so the reparametrization mode must be treated quantum mechanically rather than semiclassically.

The exponentially large scale in the table is a prediction obtained by extrapolating a smooth spectral density. It is not a proved local-spacing law for finite-N SYK. The parameter E_0 in (2.14) is not a sample ground-state eigenvalue. The Schwarzian density near the lower spectral edge E_0 has the square-root form

$$\rho(E_0 + \varepsilon) \asymp e^{S_0} C^{3/2} \varepsilon^{1/2}, \qquad \varepsilon \downarrow 0, \tag{2.15}$$

up to constants independent of ε. Consequently, the smooth density predicts

$$\int_0^{\varepsilon} \rho(E_0 + u)\, du \asymp e^{S_0} C^{3/2} \varepsilon^{3/2}.$$

The continuum approximation reaches the first-level scale when this quantity is of order one. Thus

$$\varepsilon_* \asymp C^{-1} e^{-2S_0/3}, \qquad \beta_{\mathrm{edge}} \asymp C e^{2S_0/3}, \tag{2.16}$$

At the level of this effective-theory estimate,

$$\lim_{N\to\infty} \frac{1}{N} \log(\beta_{\mathrm{edge}} J) = \frac{2}{3} s_0(q). \tag{2.17}$$

For $q = 4$ the exponential rate is approximately 0.1549. Polynomial prefactors depend on conventions and on the observable used to locate the crossover; the exponent $2s_0/3$ follows from the assumed square-root edge. It combines the exponential number e^{S_0} of low-energy states with the square-root spectral edge. More generally, an edge law $\rho(E_0 + \varepsilon) \asymp e^{S_0} \varepsilon^{\alpha}$ would give the exponential rate $s_0/(\alpha + 1)$.

The order-of-limits distinction in (2.11) is therefore a statement about spectral resolution. At fixed β, the large-N limit cannot distinguish individual low-lying eigenvalues. Within the smooth approximation, the number of levels in the quantum-Schwarzian thermal window is of order

$$\int_0^{1/\beta} \rho_{\mathrm{smooth}}(E_0 + \varepsilon)\, d\varepsilon \asymp e^{S_0} \left(\frac{C}{\beta}\right)^{3/2}, \qquad C \ll \beta.$$

It remains large when $C \ll \beta \ll \beta_{\text{edge}}$. Thus order-one fluctuations of the soft mode occur at a polynomial scale, whereas individual edge levels are predicted to become visible at an exponential scale.

An assumption such as $\liminf N^{-1}\log(\beta_N J) > 0$ alone does not locate the discrete-spectrum regime. Its exponential rate must be compared with $2s_0(q)/3$, and its prefactor matters at equality. For a fixed sample, write the distinct energies as $\mathcal{E}_0 < \mathcal{E}_1 < \cdots$, with multiplicities $d_0, d_1, \ldots$. The exact identity

$$Z_g(\beta) = d_0 e^{-\beta\mathcal{E}_0}\left(1 + \sum_{a\geq 1}\frac{d_a}{d_0}e^{-\beta(\mathcal{E}_a-\mathcal{E}_0)}\right)$$

shows that ground-state dominance requires the remaining sum to be small. In a joint limit, even this condition needs control of both gaps and multiplicities. The smooth density predicts a candidate scale; it does not prove that condition for typical disorder. Annealed quantities such as $\log \mathbb{E}Z$ require additional control of rare samples [13, 14].

2.5. The four-point ladder kernel

The two-point function determines the mean field. The four-point function probes interactions between collective excitations and is the gateway to the operator spectrum and chaos.

Connected four-point function. A ladder graph consists of two chains of propagators joined by successive rungs; the kernel K inserts one rung. Write $\tau_{ab} = \tau_a - \tau_b$. The resolvent identity below is rigorous for a bounded operator on a Banach space with $\|K\| < 1$: if $S_m = \sum_{j=0}^{m} K^j$, then $(I-K)S_m = I - K^{m+1}$, and $K^{m+1} \to 0$ in norm. Beyond this sufficient condition the function space and spectral prescription are essential; an eigenvalue one obstructs inversion.

For distinct flavor indices in an averaged singlet channel, a normalized connected four-point function has size $\mathcal{F} = O(N^{-1})$. Its leading diagrams form a ladder series. If K is the integral operator adding one rung, then

$$\mathcal{F} = \mathcal{F}_0 + K\mathcal{F}_0 + K^2\mathcal{F}_0 + \cdots = (I-K)^{-1}\mathcal{F}_0, \tag{2.18}$$

whenever the series is interpreted in a suitable spectral sense.

For quartic Majorana SYK, the conformal kernel can be written schematically as

$$K(\tau_1,\tau_2;\tau_3,\tau_4) = -3G(\tau_{13})G(\tau_{24})G(\tau_{34})^2,$$

with convention-dependent signs.

Conformal diagonalization. A conformal primary of dimension h transforms as $\mathcal{O}(\tau) \mapsto f'(\tau)^h \mathcal{O}(f(\tau))$ under the projective group. Its derivatives are descendants. A fermion bilinear contains two fermionic fields, possibly with time derivatives and singlet flavor sums; its dimension need not be twice the fermion dimension. An operator product expansion (OPE) is a short-distance expansion inside correlators, not an equality of bounded matrices in norm. The coefficients C_n are OPE coefficients. Details omitted: construction of the conformal eigenfunctions, evaluation of their kernel integrals, and completeness of the resulting spectral representation.

In the conformal limit, K commutes with the $SL(2,\mathbb{R})$ action. Hence it can be diagonalized using one-dimensional conformal eigenfunctions. For the $q = 4$ model, a commonly used eigenvalue

function is

$$k(h) = -\frac{3}{2}\frac{\tan\left(\frac{\pi}{2}(h-\frac{1}{2})\right)}{h-\frac{1}{2}}. \tag{2.19}$$

The dimensions h of fermion bilinear operators satisfy

$$k(h) = 1. \tag{2.20}$$

The special solution $h = 2$ corresponds to the reparametrization soft mode.

The substitution is elementary: $h - 1/2 = 3/2$, $\tan(3\pi/4) = -1$, and $k(2) = 1$. Also direct differentiation gives $k'(2) = -\pi - 2/3 \neq 0$. The equation alone does not prove that every root is a physical operator dimension; that interpretation uses the omitted spectral and OPE analysis. The strict conformal kernel has eigenvalue one on this sector. The kinetic correction lifts the degeneracy. Inverting $I - K$ enhances the corresponding response: for thermal-time separations, normalized soft contributions can scale as $\beta/C \asymp \beta J/N$, rather than uniformly as $1/N$. This is the same fluctuation parameter as in (2.10).

Interpretation as an operator-spectrum problem. In a conformal theory one expects an operator product expansion of the schematic form

$$\psi_i(\tau_1)\psi_i(\tau_2) \sim \sum_n C_n |\tau_{12}|^{h_n - 2\Delta} \mathcal{O}_{h_n}(\tau_2).$$

The transcendental equation (2.20) determines the tower h_n. Thus a nontrivial interacting spectrum is reduced to the spectral analysis of an explicit integral operator.

This reduces the formal calculation of bilinear dimensions to an integral-operator eigenvalue problem [4]. Its domain and the treatment of the $h = 2$ sector must be specified before interpreting the resolvent analytically.

3. Quantum chaos and the absence of quasiparticles

3.1. Quantum chaos

Operator growth. The support of an operator in a Majorana basis is the set of monomials with nonzero coefficients; operator growth refers to weight developing on monomials of higher degree. An out-of-time-order correlator (OTOC) is a product whose sequence of real-time arguments is not time ordered, such as $t, 0, t, 0$. The factors $\rho^{1/4}$ below separate the insertions in imaginary time. For odd fermionic probes use the graded commutator $AB - (-1)^{|A||B|}BA$, hence the anticommutator when both probes are odd.

A quantum many-body system has no classical phase-space trajectory in the usual sense. To measure sensitivity to initially simple perturbations, one studies operator growth. Let $W(t) = e^{iHt}We^{-iHt}$. If W and V initially act on effectively independent degrees of freedom, then $[W(0), V]$ may be small or vanish. Under chaotic evolution, $W(t)$ becomes increasingly nonlocal in operator space, and the commutator grows.

For self-adjoint even operators W and V, a nonnegative diagnostic is $C(t) = -\langle [W(t), V]^2 \rangle_\beta$.

Indeed $B = [W(t), V]$ satisfies $B^* = -B$, so $-\operatorname{Tr}\left(\rho B^2\right) = \operatorname{Tr}(\rho B^* B) \geq 0$. Moreover $C(t) \leq 4\|W\|^2\|V\|^2$, since $\|[W(t), V]\| \leq 2\|W\|\|V\|$. Thus unbounded exponential growth cannot persist at

fixed N. A more analytically controlled object is a regularized OTOC such as

$$F(t) = \mathrm{Tr}\left(\rho^{1/4} W(t) \rho^{1/4} V \rho^{1/4} W(t) \rho^{1/4} V\right).$$

Exponential growth. Here F_{disc} is the factorized leading contribution, λ_L is the growth rate of the small connected correction, and the scrambling time is when that correction becomes comparable with the leading term. This use of "Lyapunov exponent" describes an asymptotic correlator, not a derivative of a classical trajectory. Algebraically, if the threshold is $B > 0$, then $Ae^{\lambda_L t_*}/N = B$ gives $t_* = \lambda_L^{-1} \log(NB/A)$. Details omitted: the retarded-kernel calculation producing the exponential and its amplitude.

In a large-N scrambling window one finds

$$F(t) = F_{\mathrm{disc}} - \frac{A}{N} e^{\lambda_L t} + \cdots . \tag{3.1}$$

In the leading strong-coupling, large-N approximation,

$$\lambda_L = \frac{2\pi}{\beta} = 2\pi T. \tag{3.2}$$

The correction becomes order one when

$$\frac{1}{N} e^{\lambda_L t_*} \sim 1,$$

so

$$t_* \sim \frac{\beta}{2\pi} \log N.$$

This estimate assumes that the prefactor A is of order one in the chosen scaling. If β grows with N, its temperature dependence must also be retained when locating the scrambling time.

The chaos bound. Maldacena, Shenker, and Stanford [7] derived, under assumptions of analyticity, factorization, and thermal regularity, the bound

$$\lambda_L \leq \frac{2\pi}{\beta}. \tag{3.3}$$

SYK saturates it in the strong-coupling large-N limit.

Details omitted: the analytic strip estimates in the chaos-bound theorem and a uniform saturation estimate for microscopic SYK; the finite-matrix positivity calculation above does not establish either.

The relevant mathematical statement is not that every finite-N OTOC equals a pure exponential. The claim is an asymptotic one with a time window: $\beta \ll t \ll \beta \log N$. A rigorous maximal-chaos theorem would need uniform control of errors in precisely such a joint limit.

Retarded ladder kernel. The exponent λ_L can be extracted from an analytically continued ladder equation. One looks for eigenfunctions of a retarded kernel with time dependence

$$f(t_1, t_2) \sim e^{\frac{\lambda}{2}(t_1 + t_2)} \phi(t_1 - t_2).$$

The leading value of λ for which the corresponding eigenvalue equals one gives the Lyapunov exponent. The same exceptional reparametrization sector that produces the $h = 2$ Euclidean mode controls the maximal real-time growth.

Thus the same sector relates three calculations:

$$\text{nearly zero Euclidean mode} \quad \leftrightarrow \quad \text{Schwarzian action} \quad \leftrightarrow \quad \text{maximal OTOC growth.}$$

3.2. Fermi liquids

The term "non-Fermi liquid" is meaningful only after the Fermi-liquid paradigm is stated carefully.

Retarded Green function near a Fermi surface. Momentum k is the Fourier variable dual to position. The dispersion ε_k is the one-particle energy measured from the chemical potential μ, the parameter conjugate to conserved particle number. A regular Fermi surface here is a smooth codimension-one zero set of $\varepsilon_k + \operatorname{Re}\Sigma^R(0,k)$ with nonzero normal derivative and vanishing damping at zero frequency. $k_\perp$ is signed displacement along its normal; v_F^* is the renormalized normal slope. The subscript "inc" denotes a less singular, incoherent remainder. The spectral function is $-\pi^{-1}\operatorname{Im} G^R(\omega + i0, k)$.

Consider an interacting translationally invariant fermion system in spatial dimension $d \geq 2$. The retarded Green function is

$$G^R(\omega, k) = \frac{1}{\omega - \varepsilon_k - \Sigma^R(\omega, k)}. \tag{3.4}$$

A Fermi surface may be characterized by a singularity or zero of the inverse Green function at zero frequency. Near a regular point k_F, let $k_\perp$ be the momentum normal to the Fermi surface.

Landau Fermi-liquid theory predicts

$$G^R(\omega, k) \simeq \frac{Z}{\omega - v_F^* k_\perp + i\Gamma(\omega, T)} + G^R_{\text{inc}}, \tag{3.5}$$

where $Z > 0$ is the quasiparticle residue.

The lifetime criterion. A pole at $z_k = E_k - i\Gamma_k$ represents a useful quasiparticle only if its width is asymptotically small compared with its energy:

$$\frac{\Gamma_k}{|E_k|} \longrightarrow 0 \quad \text{as } k \to k_F. \tag{3.6}$$

For an ordinary Fermi liquid, $\Gamma(\omega, T) \sim c_1\omega^2 + c_2T^2$ up to dimension-dependent logarithms. Hence $\Gamma(\omega, 0) = o(|\omega|)$. This is the core mathematical content of the statement that low-energy quasiparticles are long lived.

The quasiparticle residue is formally

$$Z = \left[1 - \partial_\omega \operatorname{Re}\Sigma^R(\omega, k_F)|_{\omega=0}\right]^{-1},$$

and also appears as the jump of the momentum distribution at the Fermi surface in a standard Fermi liquid.

The Taylor expansion giving the residue and the pole lifetime is verified in Appendix A.9. Details omitted: the interaction-dependent decay-rate law and the momentum-distribution jump theorem.

Definition 3.1 (Quasiparticle-like singularity). A low-energy state is quasiparticle-like at k_F if there exist $Z_* > 0$, $v_F^* \neq 0$, and $\Gamma(\omega) = o(|\omega|)$ such that the most singular part of G^R has the form

$$\frac{Z_*}{\omega - v_F^* k_\perp + i\Gamma(\omega)}.$$

This definition captures the aspect of Fermi-liquid theory most relevant to SYK: the existence of an asymptotically sharp one-particle pole.

3.3. SYK and non-Fermi-liquid behavior

A Green-function formulation. The label "non-Fermi liquid" covers several mechanisms. A useful sufficient signal is that the self-energy dominates the bare frequency in the infrared. Suppose $\Sigma^R(\omega) \sim c(-i\omega)^\alpha$, $0 < \alpha < 1$. Then

$$\frac{|\Sigma(\omega)|}{|\omega|} \sim |\omega|^{\alpha-1} \to \infty.$$

The interaction correction is larger than the bare kinetic term, and a narrow quasiparticle pole cannot be treated perturbatively.

SYK scaling. For quartic SYK,

$$G(\tau) \sim \frac{\operatorname{sgn}\tau}{|\tau|^{1/2}}.$$

Fourier transformation gives, schematically, $G(i\omega) \sim i\,\operatorname{sgn}(\omega)|\omega|^{-1/2}$. Since in the infrared $G^{-1} \simeq -\Sigma$, we obtain $\Sigma(i\omega) \sim i\,\operatorname{sgn}(\omega)|\omega|^{1/2}$. Therefore

$$\frac{|\Sigma(\omega)|}{|\omega|} \sim |\omega|^{-1/2} \to \infty.$$

The self-energy therefore dominates the bare frequency in this approximation.

Another way to say the same thing is that the retarded Green function has a branch-type power-law singularity rather than a simple isolated quasiparticle pole: $G^R(\omega) \sim (\omega + i0^+)^{-1/2}$ up to a complex phase fixed by causality and particle-hole symmetry.

The meaning of non-Fermi liquid. The basic SYK model is zero-dimensional in space. It has no momentum and therefore no literal Fermi surface. It is more precise to call the infrared state a local non-Fermi-liquid fixed point or a non-quasiparticle quantum liquid. To describe an actual metal one must add charge and spatial structure.

Complex SYK. A conserved charge is a self-adjoint Q with $[H, Q] = 0$; $e^{i\theta Q}$ is its $U(1)$ symmetry. The grand-canonical Gibbs weight is $e^{-\beta(H_{\text{int}} - \mu Q)}$. Here $[Q, c_i^\dagger] = c_i^\dagger$ and $[Q, c_i] = -c_i$, so $[Q, c_i^\dagger c_j^\dagger c_k c_l] = (1 + 1 - 1 - 1)c_i^\dagger c_j^\dagger c_k c_l = 0$. Taking the adjoint reverses both antisymmetric pairs, producing two minus signs, so the stated condition on $g_{ij;kl}$ makes the interaction self-adjoint.

Introduce complex fermions $c_i, c_i^\dagger$ and a conserved charge

$$Q = \sum_i c_i^\dagger c_i.$$

A representative Hamiltonian, with couplings antisymmetric in each pair and satisfying $g_{ij;kl} = \overline{g_{kl;ij}}$, is

$$H = \sum_{i,j,k,l} g_{ij;kl} c_i^\dagger c_j^\dagger c_k c_l - \mu \sum_i c_i^\dagger c_i.$$

In this paragraph only, use the conventional complex-fermion function $G(\tau) = -\langle T_\tau c_i(\tau) c_i^\dagger(0)\rangle$. With coupling variance scaled as U^2/N^3 and combinatorial constants absorbed into U, the large-N equations take the form

$$G(i\omega_n) = \frac{1}{i\omega_n + \mu - \Sigma(i\omega_n)},$$

$\Sigma(\tau) = -U^2 G(\tau)^2 G(-\tau)$, together with a density constraint [10]. The preceding Majorana sections use the opposite Euclidean sign convention.

SYK lattices and strange metals. One can put SYK dots on a lattice, $x \in \Lambda$, with random interactions within each site and hopping between sites. A schematic Euclidean action is

$$L = \sum_{x,i} c_{i,x}^\dagger (\partial_\tau - \mu) c_{i,x} - \sum_{\langle xx'\rangle, ij} t_{ij,xx'} c_{i,x}^\dagger c_{j,x'} - \sum_{x,ijkl} g_{ijkl,x} c_{i,x}^\dagger c_{j,x}^\dagger c_{k,x} c_{l,x}.$$

Such models can have incoherent transport without a quasiparticle Boltzmann equation.

A "strange metal" is a phenomenological term, often associated with a broad regime of approximately linear resistivity, $\rho(T) = \rho_0 + AT$, and relaxation times of order the thermal time

$$\tau_{\mathrm{rel}} \sim \frac{\hbar}{k_B T}.$$

These are related to, but not synonymous with, non-Fermi-liquid behavior.

3.4. Transport and spatial extensions

The physical importance of SYK is not that ordinary materials literally contain all-to-all Gaussian random couplings. Its importance is that it gives a controlled non-quasiparticle fixed point which can be embedded into more realistic constructions.

A solvable strongly interacting fixed point. Most analytic many-body calculations start from weakly interacting particles. SYK instead starts from a strongly interacting limit in which $|\Sigma(\omega)| \gg |\omega|$. The low-energy Green function and scaling dimension are still calculable. Thus it provides a rare explicit model of strong-coupling universality.

Transport beyond kinetic theory. Hopping is a quadratic term that transfers a fermion between sites; a dot is one spatial site with many internal modes. In the Boltzmann equation, $f(t, x, k)$ is an occupation distribution, v a velocity, and I_{coll} a collision operator. Conductivity is the coefficient relating induced electric current to an applied electric field in linear response; resistivity is its inverse in a scalar channel. A Kubo formula expresses such a coefficient through an equilibrium current correlator. For an even current J, the retarded correlator uses a commutator, $G_{JJ}^R(t) = -i\theta(t)\langle[J(t), J(0)]\rangle$, unlike the odd-fermion propagator. Its contact term is the instantaneous

response from the explicit field dependence of the current. Details omitted: model-specific transport calculations; the displayed Kubo expression is schematic.

Conventional transport is often derived from a Boltzmann equation $\partial_t f + v \cdot \nabla_x f = I_{\mathrm{coll}}[f]$. This presupposes a distribution of long-lived quasiparticles. In an SYK-derived incoherent metal, conductivity must instead be obtained directly from current-current correlation functions using a Kubo formula,

$$\sigma(\omega) = \frac{1}{i\omega} G^R_{JJ}(\omega, k = 0)$$

with the usual contact-term qualifications. This makes SYK a laboratory for transport when quasiparticle kinetic theory is unavailable.

Hydrodynamics without quasiparticles. Hydrodynamics describes slowly varying conserved densities. For example, conservation $\partial_t \varepsilon + \nabla \cdot j_E = 0$ and the constitutive assumption $j_E = -D_E \nabla \varepsilon$ give $\partial_t \varepsilon = D_E \nabla^2 \varepsilon$. A Fourier mode then decays as $e^{-D_E |k|^2 t}$. $D_E > 0$ is the energy diffusion constant. A butterfly velocity v_B is the speed of a spatial OTOC front, when such a front exists. Local criticality means scale-invariant temporal correlations without a corresponding ordinary quasiparticle dispersion. Details omitted: deriving these constitutive assumptions from a specified spatial SYK model.

Even without quasiparticles, conserved quantities can produce hydrodynamics. In spatial SYK models, energy and charge can diffuse: $\partial_t \varepsilon = D_E \nabla^2 \varepsilon + \cdots$. The coexistence of local quantum criticality, diffusion, and chaos provides a controlled setting in which to study relations among diffusion constants, butterfly velocities, and thermal time scales.

3.5. Random-matrix statistics

SYK is a structured random matrix, not a Wigner matrix. The distinction matters.

Global versus local spectral questions. Let the eigenvalues be $E_1 \le \cdots \le E_D$ with $D = 2^{N/2}$. The empirical measure is

$$\mu_N = \frac{1}{D} \sum_{a=1}^{D} \delta_{E_a / s_N}$$

for a suitable scale s_N. Global convergence of μ_N is a different question from local spacing statistics.

Symmetry sectors and local statistics. An antiunitary $\mathcal{T}$ is a conjugate-linear isometry; it is a Hamiltonian symmetry when $\mathcal{T} H \mathcal{T}^{-1} = H$. GOE is the orthogonally invariant Gaussian law on real symmetric matrices, GUE the unitarily invariant Gaussian law on complex Hermitian matrices, and GSE the symplectically invariant Gaussian law on quaternionic Hermitian matrices. Variance conventions set their energy scale. A Kramers pair consists of orthogonal eigenvectors $v, \mathcal{T} v$ at the same energy when $\mathcal{T}^2 = -I$. Appendix A.10 proves the symmetry table; it does not prove an SYK random-matrix limit. For quartic Majorana SYK, the antiunitary symmetries give the following random-matrix classes within a fixed fermion-parity sector [13]:

N mod 8	class	symmetry qualification
0	GOE	antiunitary symmetry squares to $+1$ within a sector
2, 6	GUE	antiunitary symmetry exchanges the parity sectors
4	GSE	antiunitary symmetry squares to -1 within a sector

GOE, GUE, and GSE denote the Gaussian orthogonal, unitary, and symplectic ensembles. For $N \equiv 4 \pmod 8$, count each Kramers pair once when computing spacings; for $N \equiv 2, 6 \pmod 8$, do not combine the paired parity spectra. Otherwise exact degeneracies create zero gaps unrelated to level repulsion. The table concerns $q = 4$; other interaction orders require their own symmetry analysis.

Fix a sector and remove these multiplicities. If E_* is a bulk energy with smooth mean density $\bar{\rho}_N(E_*) > 0$, the locally unfolded energies are

$$x_a = \bar{\rho}_N(E_*)(E_a - E_*). \tag{3.7}$$

This normalization gives mean spacing of order one when the density varies negligibly across the window. The random-matrix prediction concerns the point process $\sum_a \delta_{x_a}$, including its gap and correlation statistics. In the three Gaussian ensembles, the bulk spacing density behaves as $p(s) \propto s^r$ for small s, with $r = 1, 2, 4$, respectively. Identifying a symmetry class does not prove convergence of the SYK point process.

Level repulsion means suppression of very small gaps; spectral rigidity means smaller fluctuations of level counts than for independent levels. Details omitted: the Gaussian-ensemble spacing asymptotics and SYK local universality.

At a spectral edge the mean density vanishes, so (3.7) is not the appropriate normalization. One instead centers the edge and rescales by a scale such as ε_* in (2.16). A square-root density alone does not determine the edge point process. Bulk and edge universality are separate questions.

Spectral form factor. Define $K(\beta, t) = \mathbb{E}\,|Z(\beta + it)|^2$. In eigenvalue coordinates,

$$K(\beta, t) = \mathbb{E} \sum_{a,b} e^{-\beta(E_a+E_b)} e^{-it(E_a-E_b)}.$$

Thus K is a Fourier transform of a two-level correlation measure. Chaotic random-matrix statistics produce the qualitative sequence

$$\text{dip} \longrightarrow \text{ramp} \longrightarrow \text{plateau}.$$

The ramp is a long-range signature of spectral rigidity.

Here dip denotes the initial decrease to a minimum, ramp an approximately increasing intermediate regime, and plateau a late-time scale set by diagonal spectral contributions. The fixed-sample time-average calculation of the plateau is given in Appendix A.10; details omitted: the disorder-averaged ramp. Resolving finitely many local gaps alone does not establish a ramp: the latter also requires control of correlations over a growing range of energies and of the time scale in the Fourier transform.

Different time scales. In a semiclassical scrambling regime, $1 \ll \beta J \ll N$, one distinguishes the thermal time β, the scrambling time, and the local Heisenberg time

$$t_H(E) \asymp \delta_N(E)^{-1}, \qquad \delta_N(E) \coloneqq \bar{\rho}_N(E)^{-1}.$$

The density here counts the levels of the chosen symmetry sector. The scrambling estimate $t_* \sim \beta(2\pi)^{-1} \log N$ further assumes an order-one OTOC prefactor, as discussed after (3.1). The

Heisenberg time depends on the energy window and is typically exponential in N in a many-body spectral region. Neither time should be identified with the polynomial Schwarzian scale C.

4. The relation to nearly-AdS_2 gravity

4.1. Geometric preliminaries

We introduce the geometric objects needed to compare the SYK soft action with the boundary action of Jackiw–Teitelboim gravity. For fuller derivations, see [6, 22].

Lorentzian manifolds. Indices μ, ν label coordinates and repeated upper and lower indices are summed. $R_{\mu\nu}$ is the Ricci tensor, $R = g^{\mu\nu}R_{\mu\nu}$ the scalar curvature, Λ the cosmological constant, G_N Newton's gravitational constant, and $T_{\mu\nu}$ the stress-energy tensor. In this gravity section G_N is not the averaged Green function from (1.8). Vacuum means $T_{\mu\nu} = 0$. Euclidean signature replaces the one negative metric direction by a positive one; the Euclidean counterpart of AdS_2 is the hyperbolic plane. Details omitted: derivation of Einstein's equation as a physical field equation.

A spacetime is a smooth manifold M equipped with a Lorentzian metric g of signature $(-, +, \ldots, +)$. The metric gives the squared interval $ds^2 = g_{\mu\nu}dx^\mu dx^\nu$. Vectors are timelike, null, or spacelike according as $g(v, v)$ is negative, zero, or positive. The Levi-Civita connection determines geodesics and the curvature tensor.

Einstein's equation is

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu}. \tag{4.1}$$

Anti-de Sitter space is a maximally symmetric vacuum solution with negative cosmological constant in dimension greater than two. The two-dimensional theory used below instead imposes curvature through a dilaton equation.

AdS_2. Two-dimensional anti-de Sitter space can be written in Poincaré coordinates as

$$ds^2 = \frac{L^2}{z^2}(-dt^2 + dz^2), \qquad z > 0. \tag{4.2}$$

Its scalar curvature is constant,

$$R = -\frac{2}{L^2}.$$

Its isometry group is locally $SL(2, \mathbb{R})$. The same group is the global conformal group of one-dimensional time.

Appendix A.11 computes the curvature and three Killing fields. Global covers and disconnected isometries require qualifications; only the local Lie group and its projective boundary action are used here. This common symmetry is the first structural hint of the SYK-AdS_2 connection.

Horizons and black holes. An event horizon is a global causal boundary: the boundary of the region from which future-directed causal curves fail to reach the asymptotic exterior. It is not a material membrane. In smooth large black holes, local curvature at the horizon may be small.

For the SYK story, charged near-extremal black holes are especially important. Extremality means that, at fixed conserved charge, the black hole reaches the minimum allowed mass and has zero Hawking temperature in the classical solution. Near extremality, the near-horizon geometry develops a long approximately AdS_2 throat.

This assertion concerns suitable charged black-hole families, not every extremal solution. A throat is a near-horizon region with this approximate product geometry. Hawking temperature is the temperature associated with a regular Euclidean horizon, and surface gravity κ fixes it by $T = \kappa/(2\pi)$ in the units used here. Details omitted: construction of those black-hole solutions and the quantum-radiation argument.

4.2. JT gravity and the Schwarzian

Why two-dimensional gravity is special. In two dimensions, the Einstein-Hilbert integral

$$\int_M \sqrt{g}\, R\, d^2x$$

is topological up to boundary terms by Gauss-Bonnet. Pure two-dimensional Einstein gravity therefore has no local propagating metric degree of freedom.

Jackiw-Teitelboim gravity introduces a scalar dilaton Φ.

A dilaton is an additional scalar field coupled to curvature; in reductions from higher dimensions it can encode the transverse area. Φ_0 is constant, while Φ is varied. The boundary term $I_{\partial M}$ supplies the extrinsic-curvature terms for fixed boundary data and a counterterm removing the cutoff divergence. In Euclidean signature a common normalization is

$$I_{JT}[g,\Phi] = -\frac{\Phi_0}{16\pi G}\int_M \sqrt{g}\, R\, d^2x \tag{4.3}$$

$$-\frac{1}{16\pi G}\int_M \sqrt{g}\,\Phi\left(R+\frac{2}{L^2}\right) d^2x + I_{\partial M}. \tag{4.4}$$

Varying with respect to Φ gives

$$R + \frac{2}{L^2} = 0.$$

Thus the Euclidean metric is locally hyperbolic, and its Lorentzian counterpart is locally AdS_2. The nontrivial degree of freedom is encoded in the dilaton and in the boundary embedding.

Boundary reparametrization. Regulate the asymptotic boundary by a curve parametrized by u. The physical boundary time is related to Poincaré time by $t = f(u)$. Solving the bulk constraints and evaluating the boundary action gives

$$I_{\mathrm{grav}}[f] = -C\int_0^\beta du\left\{\tan\left(\frac{\pi f(u)}{\beta}\right), u\right\}. \tag{4.5}$$

This is the same functional form as the SYK soft action (2.9).

Appendix A.11 verifies the regulated boundary calculation with explicit boundary conditions. Details omitted: a microscopic SYK-to-gravity convergence theorem.

The safest mathematical statement of the low-energy duality is therefore $I_{\mathrm{SYK,soft}}[f] \cong I_{\mathrm{JT,boundary}}[f]$ after matching the coefficient C to $N\alpha_S$.

Symmetry interpretation. On the SYK side, the conformal equations possess approximate $\mathrm{Diff}(S^1)$ covariance, but a saddle preserves only $PSL(2,\mathbb{R})$ and the UV kinetic term explicitly breaks the larger symmetry.

On the gravity side, large boundary reparametrizations are physical except for the exact $PSL(2,\mathbb{R})$ isometries of AdS_2. The Schwarzian measures the action cost of these boundary modes.

Thus the matching is not merely numerical. It is a matching of symmetry-breaking structures: $\mathrm{Diff}(S^1)/PSL(2,\mathbb{R})$ appears on both sides.

4.3. Black-hole thermodynamics

Bekenstein-Hawking entropy. Here $\mathcal{A}_H$ is the area of a spatial horizon cross-section, c is the speed of light, and the subscript BH means black hole. The area law is an input from semiclassical gravity; details omitted: its derivation. The linear near-extremal entropy law below follows conditionally from a Schwarzian partition function with $C > 0$, not from the area formula without specifying a family of solutions.

For a semiclassical black hole,

$$S_{BH} = \frac{k_B c^3}{4G_N \hbar}\mathcal{A}_H.$$

In units $c = \hbar = k_B = 1$,

$$S_{BH} = \frac{\mathcal{A}_H}{4G_N}. \tag{4.6}$$

For a near-extremal charged black hole at fixed charge,

$$S_{BH}(T) = S_0 + \gamma_{BH} T + O(T^2). \tag{4.7}$$

Compare with large-N SYK, $S_{SYK}(T) = Ns_0 + N\gamma_{SYK}T + \cdots$. The functional form is identical.

This thermodynamic parallel predates the detailed Schwarzian matching and was an important clue that the same low-energy effective theory might be present.

Residual entropy and order of limits. On the black-hole side, S_0 is proportional to the extremal horizon area. On the SYK side, Ns_0 is the large-N zero-temperature entropy obtained when $N \to \infty$ is taken before $T \to 0$.

The correspondence should not be interpreted as an exponentially degenerate exact finite-N SYK ground state. Instead, the density of low-energy states becomes exponentially large in a scaling window whose spacing is invisible before taking the thermodynamic limit.

Schwarzian density of states. After matching C, the JT boundary theory gives the same Laplace density (2.13) and partition function (2.12). Restoring the edge energy and entropy gives (2.14). This compares smooth low-energy thermodynamics; microscopic level correlations require information beyond this density.

4.4. The holographic dictionary

It is helpful to phrase holography as an inverse problem for correlation data.

Boundary dimensions and bulk masses. Bulk means the two-dimensional interior, and boundary means its one-dimensional asymptotic time coordinate. A bulk scalar of mass m satisfies the linear Klein–Gordon equation $(\Box_g - m^2)\phi = 0$, where $\Box_g = |\det g|^{-1/2}\partial_\mu(|\det g|^{1/2}g^{\mu\nu}\partial_\nu)$. Its near-boundary powers give the following relation, proved in Appendix A.11. It is an indicial equation; the choice of admissible boundary conditions is additional data.

A scalar field of mass m in AdS_2 is related to a boundary operator of dimension h by

$$m^2L^2 = h(h-1). \tag{4.8}$$

SYK provides a tower of bilinear dimensions h_n through $k(h_n) = 1$. One may therefore assign a candidate bulk field ϕ_n with $m_n^2L^2 = h_n(h_n - 1)$. A schematic bulk action is

$$\mathcal{L}_{\text{bulk}} = \sum_n \left[\frac{1}{2}(\partial\phi_n)^2 + \frac{1}{2}m_n^2\phi_n^2\right] + \frac{1}{\sqrt{N}}\sum_{nmk}\lambda_{nmk}\phi_n\phi_m\phi_k + \cdots .$$

Boundary three-point coefficients determine the cubic couplings after known AdS_2 normalization factors are taken into account.

A cubic coupling is the coefficient of a product of three bulk fields. Details omitted: evaluation of the bulk integrals and proof that a local bulk reconstruction exists. The displayed action is a candidate organization of the correlation data.

What is established and what is not. The robust statement is that the universal low-energy reparametrization sector of SYK is described by the same Schwarzian action as nearly-AdS_2 JT gravity. This yields matching low-temperature thermodynamics and matching soft contributions to correlation functions and chaos.

A much stronger statement would be an exact microscopic equivalence $\mathcal{H}_{\text{SYK}} \cong \mathcal{H}_{\text{pure JT}}$ with equality of all finite-N observables. That is not what has been established. Ordinary SYK also contains an infinite tower of order-one-dimension bilinear operators, so pure JT gravity captures only the universal soft sector, not all microscopic data.

Remark 4.1 (A proposed formulation of effective holographic matching). For a specified class of low-energy sources g_∂ and a prescribed joint limit $N \to \infty$, $\beta \to \infty$, one would seek a normalization in which the connected generating functional $W = \log Z$ of the soft sector satisfies

$$W_{\text{SYK}}^{\text{soft}}[g_\partial] - W_{JT}^{\text{boundary}}[g_\partial] \to 0$$

with controlled errors. A source is an external function coupled to an observable in the action; derivatives of W with respect to sources generate connected correlators. This is a proposed target, not a theorem established here.

To turn this template into a precise conjecture, one must specify the sources, normalization, joint limit, and topology of convergence.

4.5. Black holes and chaos

Near a black-hole horizon, a small perturbation sent in at an early time undergoes an exponentially large relative boost compared with a later probe. In the semiclassical shock-wave calculation this produces an exponential effect with scale

determined by the horizon surface gravity. A shock wave is the idealized metric perturbation of a highly boosted localized pulse. Details omitted: the scattering calculation. The kinematic boost relation is verified in Appendix A.11. Its proposed OTOC rate is

$$\lambda_L = \frac{2\pi}{\beta}.$$

The same exponent appears from the Schwarzian mode and from the SYK retarded ladder kernel.

Thus the triangle

$$\text{SYK} \quad \longleftrightarrow \quad \text{Schwarzian} \quad \longleftrightarrow \quad \text{near-extremal black hole}$$

explains the common value of the chaos exponent. The common effective reparametrization theory accounts for this agreement.

Restoring constants,

$$\lambda_L = \frac{2\pi k_B T}{\hbar},$$

so the Lyapunov time is

$$\tau_L = \frac{\hbar}{2\pi k_B T}.$$

The same thermal time also appears in discussions of Planckian relaxation in strange metals, although the equality of all such time scales is not a universal theorem.

4.6. Complex SYK and charge

The charged complex SYK model has an additional parameter: the conserved density

$$\mathcal{Q} = \frac{1}{N}\langle Q \rangle.$$

At nonzero density, the low-frequency spectral function need not be particle-hole symmetric. The conformal Green function contains a spectral-asymmetry parameter, often denoted $\mathcal{E}$ or an equivalent angle.

Particle-hole symmetry means equal particle-addition and particle-removal spectral weights after reversing frequency. One possible convention is $e^{-2\pi\mathcal{E}} = a_+/a_-$ when $G(\tau) \sim -a_+\tau^{-2\Delta}$ for $\tau > 0$ and $G(\tau) \sim a_-(-\tau)^{-2\Delta}$ for $\tau < 0$, with $a_\pm > 0$. This defines a logarithmic amplitude ratio; reversing the convention reverses its sign. The charge response is the derivative of the density with respect to μ.

A notable thermodynamic relation links spectral asymmetry to the charge derivative of the zero-temperature entropy:

$$\frac{\partial s_0}{\partial \mathcal{Q}} \propto \mathcal{E}$$

with convention-dependent proportionality constants.

Details omitted: the thermodynamic derivation and its conversion to a chosen spectral-asymmetry convention. An analogous relation appears in the thermodynamics of charged near-extremal black holes with an AdS_2 throat.

The comparison therefore involves charge response and spectral asymmetry in addition to the entropy [8, 9].

5. Probabilistic results and further questions

5.1. The probabilistic formulation

We now recast SYK in a language closer to random operators and mean-field disordered systems.

A sequence of random operator models. For each even N, let $(\Omega_N, \mathscr{F}_N, \mathbb{P}_N)$ carry independent standard normal variables g_I. Let $\mathcal{H}_N \cong \mathbb{C}^{2^{N/2}}$ and define

$$H_N(\omega) = \sigma_N \sum_{I \in \mathcal{I}_N} g_I(\omega)\psi_I,$$

where $\psi_I = i^{q/2}\psi_{i_1}\cdots\psi_{i_q}$ and $\sigma_N^2 = (q-1)!/N^{q-1}$. The quenched pressure is

$$p_N(\beta) = \frac{1}{N}\mathbb{E}\log \operatorname{Tr} e^{-\beta H_N}.$$

A basic mathematical problem is to prove existence and characterize $p(\beta) = \lim_{N\to\infty} p_N(\beta)$. Gamarnik, Pernice, Schmidhuber, and Zlokapa [18] establish the annealed and quenched limits at sufficiently high fixed temperature. The September 2026 revision of He [19] reports convergence to the Schwinger–Dyson pressure for every fixed $\beta > 0$ and fixed even $q \geq 4$, as well as the limiting spectral edge. These thermodynamic statements do not themselves supply uniform infrared or real-time correlation estimates.

Details omitted: the proofs of these cited limit theorems; their statements are reported from the references, not independently established in these notes.

Fluctuations at fixed temperature. Write $P_N(\beta, g) = N^{-1}\log Z_g(\beta)$. Beyond its deterministic limit, one may ask for a normalization a_N and a nondegenerate limit

$$a_N\big(P_N(\beta, g) - \mathbb{E}P_N(\beta, g)\big) \Longrightarrow X_\beta.$$

Gaussian concentration supplies an upper bound on variance, but neither determines a_N nor establishes a Gaussian limit. The small-temperature-parameter expansion provides a useful check on possible scales.

Example 5.1 (The first random Taylor coefficient). Let $m_N = \binom{N}{q}$ and $\tau_N = 2^{-N/2}\operatorname{Tr}$. Since $\tau_N(H_q) = 0$ and the Clifford monomials are orthogonal,

$$P_N(0, g) = \tfrac{1}{2}\log 2, \qquad \partial_\beta P_N(0, g) = 0, \qquad \partial_\beta^2 P_N(0, g) = \frac{\sigma_N^2}{N}\sum_I g_I^2.$$

Consequently the centered second derivative Y_N satisfies

$$Y_N \coloneqq \partial_\beta^2 P_N(0, g) - \mathbb{E}\partial_\beta^2 P_N(0, g), \qquad \operatorname{Var}(Y_N) = \frac{2\sigma_N^4 m_N}{N^2} \sim \frac{2(q-1)!}{q}N^{-q}.$$

Moreover,

$$\frac{NY_N}{\sigma_N^2\sqrt{2m_N}} = \frac{1}{\sqrt{2m_N}}\sum_I (g_I^2 - 1) \Longrightarrow \mathcal{N}(0, 1).$$

This is the ordinary central limit theorem for independent squared Gaussians. It concerns a Taylor coefficient at $\beta = 0$, not the pressure at fixed positive β. Passing to the latter requires control of all remaining coefficients uniformly in N.

The annealed expansion

$$\mathbb{E}Z_g(\beta) = \sum_{m=0}^{\infty} \frac{(-\beta)^m}{m!} \mathbb{E}\,\mathrm{Tr}\, H_q^m$$

has only even terms by symmetry of the disorder law. At fixed N, Gaussian integrability justifies the expansion. Its terms involve Gaussian pairings and Clifford signs. A fluctuation proof could use such expansions, sharper estimates on derivatives of P_N, or martingale differences obtained by revealing the couplings. In each approach, uniformity in β is a separate issue.

A dynamical order parameter. The large-N prediction is that

$$G_\beta(\tau) = \lim_{N\to\infty} \frac{1}{N} \sum_i \mathbb{E}\langle T_\tau \psi_i(\tau)\psi_i(0)\rangle$$

is deterministic and solves

$$G^{-1}(i\omega_n) = -\frac{i\omega_n}{2} - \Sigma(i\omega_n),$$
$$\Sigma(\tau) = G(\tau)^3.$$

This resembles a mean-field fixed-point equation, but the order parameter is a function rather than a scalar.

Cavity intuition. The cavity method separates one degree of freedom from the rest; the latter form its bath. A memory kernel relates the present response to earlier times by integration, in contrast with a time-local coefficient. The algebraic decomposition below is exact. Details omitted: estimates proving deterministic bath correlations and the self-consistent limit.

For the quartic model, separate one Majorana while keeping the N-dependent normalization fixed: $H_N = H_N^{(0)} + \psi_N B_N$, where

$$B_N = \sigma_N \sum_{i<j<k<N} g_{ijkN}\psi_i\psi_j\psi_k,$$

and $H_N^{(0)}$ is the sum of the terms whose index sets do not contain N. Notice that $H_N^{(0)}$ uses σ_N, so it is not literally the normalized $(N-1)$-fermion Hamiltonian. The distinguished mode ψ_N interacts with a random dynamical bath generated by the remaining $N-1$ modes. In the large-N limit, the bath's two-time correlations become self-consistent and deterministic. The resulting memory kernel is precisely the self-energy $\Sigma(\tau) = G(\tau)^3$. This is the noncommutative, dynamical analogue of a cavity field.

Replica method. A replica is a separate copy of the system with the same disorder realization. Integer n copies have tensor-product Hilbert space $\mathcal{H}^{\otimes n}$ and partition function Z_g^n. A replica-diagonal saddle has $G^{ab} = \delta_{ab}G$, with a, b the copy labels. Replica symmetry breaking refers to a saddle structure not described by this simple symmetric ansatz. Spin-glass order is persistent random ordering often diagnosed by a nontrivial overlap distribution; the Parisi description organizes such overlaps hierarchically. Details omitted: selecting the SYK replica saddle or excluding spin-glass order in any particular regime.

The quenched average contains $\mathbb{E}\log Z$. Physicists often use

$$\mathbb{E}\log Z = \lim_{n\to 0} \frac{\mathbb{E}Z^n - 1}{n}. \tag{5.1}$$

For positive integers n, Z^n introduces n replicas. One computes $\mathbb{E}Z^n$, analyzes saddle points, and then analytically continues in n. This procedure is not automatically rigorous because the analytic continuation from integers to $n = 0$ is not unique without additional control.

At fixed N, the derivative identity itself is rigorous for real n near zero, as shown in Appendix A.7. The unproved operation is recovering that derivative from an integer-replica saddle calculation and interchanging it with $N \to \infty$.

In ordinary SYK the standard saddle is replica diagonal in the usual parameter regime, unlike the low-temperature Parisi structure of the Sherrington-Kirkpatrick spin glass. The contrast is physically important: random couplings do not imply spin-glass order by themselves. Fermionic anticommutation changes the structure of frustration and low-energy states.

5.2. Scope of the available results

A useful map of the subject is the following.

Finite-N mathematics. At finite N, the following objects are completely well-defined:

$$H_N, \quad Z_N(\beta), \quad \rho_{\beta,N}, \quad G_N(\tau), \quad \mathrm{spec}(H_N), \quad \mathrm{OTOC}_N(t).$$

They are finite-dimensional random operator quantities.

Rigorous large-N results. Relevant rigorous results include global spectral laws [16], combinatorial melonic dominance [15], and free-energy limits [18, 19]. Proposition 1.4 supplies an elementary concentration bound in the present normalization. The exact hypotheses and normalizations vary among papers. These results should be viewed as mathematical foundations for parts of the standard physics picture, not as a single theorem proving every infrared claim.

Controlled but formal physics. The chain

$$H_N \xrightarrow{N\to\infty} (G,\Sigma) \xrightarrow{\omega\to 0} G_c \xrightarrow{\text{soft fluctuations}} I_{\mathrm{Sch}} \xrightarrow{\text{real time}} \lambda_L = 2\pi/\beta$$

is supported by mutually consistent diagrammatic, saddle-point, numerical, and effective-field-theory calculations. But obtaining all arrows as one rigorous theorem with uniform error bounds in a joint low-temperature and large-N limit remains a substantial challenge.

The holographic layer. The equality of the Schwarzian effective action on the SYK and JT sides is a concrete calculational correspondence. The interpretation as a full microscopic quantum-gravity dual is more ambitious and should be stated separately.

5.3. Further questions

Every item in this subsection is a proposed research target. Details are omitted because no proof of these targets is supplied here.

The model generates a long list of precise questions.

Derive the bilocal variational principle directly. Beyond the pressure limit, derive the generating functional with time-dependent sources directly from the finite-N Gibbs state. Specify a regularized collective action and prove convergence to the physically selected Schwinger–Dyson saddle, including its source derivatives. This requires control of saddle selection and correlation functions. An unrestricted infimum over G and Σ is not justified: the multiplier contour and the logarithmic determinant need not define a real minimization problem.

Rigorous infrared scaling. For $q = 4$, take $\beta_N \to \infty$ with $\beta_N/N \to 0$, and write $G_{N,\beta}$ for the averaged correlator in (1.8), with its temperature dependence explicit. For every fixed $0 < a < 1/2$, a possible target is

$$\sup_{u\in[a,1-a]} \left| \frac{G_{N,\beta_N}(\beta_N u)}{G_{\beta_N,c}(\beta_N u)} - 1 \right| \longrightarrow 0. \tag{5.2}$$

The relative normalization matters because both correlators tend to zero. An error estimate should distinguish the kinetic correction, which decreases at low frequency, from soft fluctuations governed by β_N/C_N. Their precise rates depend on the observable and norm; the scale comparison alone does not prove an $O(\beta_N^{-1}) + O(\beta_N/N)$ bound. A theorem for times approaching the endpoints would require additional ultraviolet control.

Derive the Schwarzian as a fluctuation theory. At $\beta_N/C_N \to \lambda \in (0,\infty)$, formula (2.10) suggests a normalized effective average with weight $e^{-\mathcal{A}[g]/\lambda}$ on the projective quotient. Specify its measure, a regularization, and observables, then prove convergence of the microscopic generating functional to those averages. One must control the remaining bilocal directions and their contribution to the coefficient. The Grassmann formulation does not supply an ordinary probability law on G or g without an additional construction.

At $\beta_N/C_N \to \infty$, the corresponding target is the quantum Schwarzian spectral representation, not a Gaussian expansion about $g(u) = u$. Correlation functions require matrix-element data as well as eigenvalue density [21]. For example, a self-adjoint finite-dimensional observable has the exact expansion

$$\langle \mathcal{O}(\tau)\mathcal{O}(0)\rangle_{\beta,g} = \frac{1}{Z_g(\beta)} \sum_{a,b} e^{-(\beta-\tau)E_a - \tau E_b} |\mathcal{O}_{ab}|^2, \qquad 0 < \tau < \beta.$$

A Schwarzian limit must therefore identify the limit of the weighted two-energy measure appearing in this expression.

Four-point convergence. Prove convergence of connected four-point functions after N rescaling:

$$N\,\mathcal{F}_N \longrightarrow (I-K)^{-1}\mathcal{F}_{0,\infty}$$

in a suitable function space, where $\mathcal{F}_{0,\infty} = \lim_{N\to\infty} N\mathcal{F}_{0,N}$ is the rescaled seed. Control the singular $h = 2$ sector by a separate perturbation theory.

A maximal-chaos theorem. Specify operators V_N, W_N, temperature scaling, and a time window such that

$$C_N(t) = \frac{a}{N} e^{2\pi t/\beta}(1 + o(1))$$

uniformly for $1 \ll t/\beta \ll \log N$. This requires precise real-time analytic continuation and error control in a regime where small errors can be exponentially magnified.

A mesoscopic spectral law. Let w be a nonnegative smooth function of compact support with $\int_{\mathbb{R}} w = 1$. Smooth the unnormalized finite-N counting measure by

$$\rho_{N,\eta}(E) \coloneqq \sum_{a=1}^{2^{N/2}} \frac{1}{\eta} w\left(\frac{E - E_a}{\eta}\right).$$

This differs from the normalized global measure μ_N. Let $C_N \asymp N$, and specify an edge centering $E_{0,N}$ and an entropy normalization $S_{0,N}$. A candidate local-law target is

$$\sup_{x\in[a,b]} \left| \frac{\rho_{N,\eta_N}(E_{0,N} + x/C_N)}{e^{S_{0,N}} C_N} - \frac{1}{2\pi^2} \sinh\left(2\pi\sqrt{2x}\right) \right| \xrightarrow{\mathbb{P}} 0, \quad 0 < a < b < \infty. \tag{5.3}$$

On this interval the predicted mean spacing is of order $e^{-S_{0,N}}/C_N$, so a necessary scale separation is

$$\frac{e^{-S_{0,N}}}{C_N} \ll \eta_N \ll \frac{1}{C_N}.$$

These inequalities express many levels per smoothing window and fine resolution on the Schwarzian energy scale. They are not sufficient by themselves to prove (5.3). The centering may require a sample-dependent correction, whose size must be controlled; a leading order-N edge theorem does not determine it to accuracy $o(C_N^{-1})$.

As x approaches zero, the local spacing changes and the edge scale (2.16) replaces the preceding estimate. Smoothing is useful for a density statement; discreteness alone does not obstruct weak convergence of suitably normalized measures.

Local spectral universality. In a fixed symmetry sector, prove convergence of the bulk process defined by (3.7) to the corresponding Gaussian-ensemble process. Formulate the edge problem separately, with its own centering and scale. Gap distributions, ratios, and finite-order correlation functions are possible observables, but convergence of one is not automatically equivalent to convergence of all. Spectral form factors require additional estimates on the range of energies and times. These questions concern the correlations left undetermined by a smooth Schwarzian density.

Universality in the disorder law. Universality means that a specified limit is unchanged across a class of coupling distributions after matching their normalization. Lindeberg replacement swaps independent inputs one at a time and estimates the accumulated change of an observable by Taylor expansion.

Replace the standard normal variables g_I by centered, unit-variance variables with sufficient moments. Ask whether the thermodynamic limit, infrared scaling, Schwarzian sector, and chaos exponent are universal. At the global spectral level this resembles Lindeberg replacement. At the level of soft modes and OTOCs it is far subtler.

Sparse SYK. A q-uniform hypergraph is a vertex set with edges that are q-element subsets. Here vertices are flavors and retained interactions are edges. Sparse means that far fewer than $\binom{N}{q}$ edges are kept. Localization refers to confinement of relevant states or dynamics to restricted degrees of freedom; a precise version depends on the observable.

Retain only a random subset of q-body interactions. The interaction structure becomes a random q-uniform hypergraph. Natural questions include the connectivity threshold for melonic mean-field

behavior, concentration, localization, and the transition from sparse-cluster physics to dense SYK universality.

Double scaling. A chord diagram places the $2m$ factors of a moment around a circle and joins paired factors by chords. Crossings encode the signs from interchanging Clifford monomials. Double scaling means taking two parameters, here N and q, to infinity with a prescribed relation; q^2/N is a common parameter to hold fixed.

Let $q = q(N)$ grow with N. Then chord diagrams and noncommutative moment methods become natural. The fixed-q and growing-q limits give different asymptotic theories.

Mathematical definition of a non-Fermi liquid. For a spatial fermionic system, formulate hypotheses under which the failure of the quasiparticle property can be stated purely in terms of singularities of $G^R(\omega, k)$. A possible criterion is the absence of any decomposition

$$G^R(\omega, k) = \frac{Z_*}{\omega - v_F^* k_\perp + i\Gamma(\omega)} + R(\omega, k)$$

with $Z_* > 0$, $\Gamma(\omega) = o(|\omega|)$, and R less singular near the Fermi surface.

Holography as an inverse problem. Given conformal data $\{h_n, C_{nmk}, C_{nmkl}, \ldots\}$, find hypotheses ensuring the existence of a local or approximately local two-dimensional bulk action reproducing the boundary correlators order by order in $1/N$. Determine when such an action is unique and how bulk locality is encoded in asymptotics of boundary operator data.

A. Supplementary calculations

A.1. Collective-field calculations

We record the coefficient and sign calculations used in the main text.

From disorder to a quartic bilocal interaction. For $q = 4$, start from the Euclidean action

$$S[\psi; g] = \frac{1}{4} \sum_i \int d\tau \, \psi_i \partial_\tau \psi_i + \sigma_N \sum_{I \in \mathcal{I}_N} g_I \int d\tau \, \psi_I.$$

The Gaussian identity $\mathbb{E} e^{-gX} = e^{X^2/2}$, $g \sim \mathcal{N}(0, 1)$, produces after disorder averaging a two-time interaction. Here $\sigma_N^2/2 = 3/N^3$ and the conversion of the sum over increasing quadruples to unrestricted flavor sums contributes $1/4!$. Thus, to leading order in N,

$$\exp\left[\frac{N}{8} \int d\tau_1 d\tau_2 G(\tau_1, \tau_2)^4\right]$$

with a positive sign in the Euclidean convention of these notes. The six exchanges needed to group the two sets of four Grassmann variables give sign $+1$.

Indeed, its coefficient is

$$\frac{\sigma_N^2}{2} \frac{N^4}{4!} = \frac{3}{N^3} \frac{N^4}{24} = \frac{N}{8}.$$

This counting is the reason for the variance scaling N^{-3} in quartic SYK and fixes the G^4 coefficient in (2.3).

From the collective action to Dyson. Write $A = \frac{1}{2}\partial_\tau - \Sigma$, and regularize the time variable so that traces are finite-dimensional. Since $\delta A = -\delta\Sigma$,

$$\delta_\Sigma\left(-\frac{1}{2}\log\det A\right) = \frac{1}{2}\operatorname{Tr}\bigl(A^{-1}\delta\Sigma\bigr) = -\frac{1}{2}\int (A^{-1})_{12}\,\delta\Sigma_{12}\,d\tau_1 d\tau_2.$$

The last equality uses antisymmetry of the kernels. Adding the variation of $\frac{1}{2}\int \Sigma G$ gives $G = A^{-1}$. Variation with respect to G gives $\Sigma = G^{q-1}$. In particular, for $q = 4$,

$$G(i\omega_n) = \frac{1}{-i\omega_n/2 - \Sigma(i\omega_n)}, \qquad \Sigma(\tau) = G(\tau)^3.$$

These are stationary equations; this calculation does not select the integration contour or prove dominance of a saddle.

From scaling to $\Delta = 1/q$. Assume under $\tau \mapsto \lambda\tau$, $G(\lambda\tau) = \lambda^{-2\Delta}G(\tau)$. Then $\Sigma(\lambda\tau) = \lambda^{-2\Delta(q-1)}\Sigma(\tau)$. The convolution has scaling dimension $[\Sigma \circ G] = 2\Delta q - 1$, because one integrates once over time. To match $\delta(\tau)$, which has dimension 1, require $2\Delta q - 1 = 1$. Hence $\Delta = 1/q$.

From reparametrizations to the Schwarzian. The strict infrared equations are invariant under (2.7). Any reparametrized conformal solution G_f has the same conformal action. The microscopic derivative term breaks this degeneracy. Locality in time, projective invariance, and the requirement of three derivatives at leading nontrivial order single out the Schwarzian derivative as the natural local functional. Thus one obtains

$$I_{\text{soft}}[f] \propto N\int \{f,\tau\}\,d\tau$$

on the line, or the thermal-circle form (2.9).

This argument explains the form of the action but not its coefficient. The coefficient α_S must be matched to the full theory, including the UV region neglected by the conformal approximation.

A.2. The infrared Fourier transform

For $0 < \alpha < 1$, the distributional Fourier transform obeys

$$\int_0^\infty e^{i\omega t}t^{-\alpha}\,dt = \Gamma(1-\alpha)e^{i\pi(1-\alpha)\operatorname{sgn}\omega/2}|\omega|^{\alpha-1}$$

where the integral is defined by inserting $e^{-\varepsilon t}$ and taking $\varepsilon \downarrow 0$. For an odd function $f(t) = \operatorname{sgn}(t)|t|^{-2\Delta}$, using $\widehat{f}(\omega) = \int_{\mathbb{R}} e^{i\omega t}f(t)\,dt$, one obtains

$$\widehat{f}(\omega) = 2i\Gamma(1-2\Delta)\cos(\pi\Delta)\operatorname{sgn}(\omega)|\omega|^{2\Delta-1}, \qquad 0 < 2\Delta < 1.$$

For $q = 4$, $2\Delta = 1/2$, hence $G(i\omega) \propto i\,\operatorname{sgn}(\omega)|\omega|^{-1/2}$. The time-domain power law thus gives the low-frequency singularity.

A.3. The Clifford trace identity

Let ψ_I and ψ_J be the Clifford monomials defined in (1.11). With normalized trace $\tau_N(A) = 2^{-N/2}\operatorname{Tr} A$, one has orthogonality $\tau_N(\psi_I\psi_J) = 0$, $I \neq J$, while $\tau_N(\psi_I^2) = 1$. To verify orthogonality, let X be a nonempty product of an even number of distinct generators and choose ψ_k among its factors. Then $\psi_k X \psi_k = -X$, so cyclicity of trace gives $\operatorname{Tr} X = 0$. For distinct I, J of the same even size, $\psi_I\psi_J$ is a scalar multiple of such a product. The diagonal identity follows from $\psi_I^2 = I$.

Consequently, for $H_q = \sum_I J_I\psi_I$ with $J_I = \sigma_N g_I$,

$$\mathbb{E}\tau_N(H_q^2) = \sum_I \mathbb{E}J_I^2 = \sigma_N^2\binom{N}{q}.$$

In the present normalization this is exactly

$$\mathbb{E}\tau_N(H_q^2) = \frac{N}{q}\prod_{r=1}^{q-1}\left(1 - \frac{r}{N}\right) \sim \frac{N}{q}.$$

This gives a direct finite-dimensional explanation for the $O(N)$ second moment of the Hamiltonian.

A.4. Projective invariance of the Schwarzian

Let

$$M(f) = \frac{af + b}{cf + d}, \qquad ad - bc \neq 0.$$

A direct differentiation gives

$$(M \circ f)' = \frac{ad - bc}{(cf + d)^2} f'.$$

Substituting the first three derivatives into (2.8), all terms depending on a, b, c, d cancel, yielding $\{M \circ f, \tau\} = \{f, \tau\}$. In particular,

$$\left\{\frac{a\tau + b}{c\tau + d}, \tau\right\} = 0.$$

This identity is the elementary reason $PSL(2, \mathbb{R})$ is the exact redundancy of the Schwarzian theory.

A.5. Euclidean and real time

For a finite-dimensional Hamiltonian with eigenvectors $|n\rangle$ and energies E_n, the thermal Euclidean correlator of fermionic operators has a Lehmann expansion. For $0 < \tau < \beta$,

$$G(\tau) = \frac{1}{Z}\sum_{m,n} e^{-\beta E_n} e^{-\tau(E_m - E_n)} \langle n|\chi|m\rangle\langle m|\chi|n\rangle.$$

The retarded function is $G^R(t) = -i\theta(t)\langle\{\chi(t), \chi(0)\}\rangle_\beta$. Both are determined by the same spectral measure, and analytic continuation connects their frequency-domain representations. The difficulty in chaos calculations is not the finite-N existence of analytic continuation, but obtaining uniform asymptotics after taking large N, low temperature, and long time.

A.6. Fock, Clifford, and Gaussian identities

Occupation numbers, parity, and irreducibility. The CAR give $n_a^2 = c_a^\dagger(1 - c_a^\dagger c_a)c_a = n_a$. On a basis wedge, n_a is one or zero according as the wedge contains e_a or not, so the n_a commute. Also $[\mathcal{N}, c_a] = -c_a$ and $[\mathcal{N}, c_a^\dagger] = c_a^\dagger$, whence conjugation by $e^{i\pi\mathcal{N}}$ changes the sign of each generator. A product of q generators has parity $(-1)^q$.

For irreducibility, the vacuum projection is $P_0 = \prod_a(1 - n_a)$. Products of creation operators, followed by P_0 and products of annihilation operators, give every matrix unit $|I\rangle\langle J|$, with an adjustable sign depending on the chosen ordering. Thus the algebra generated by the $c_a, c_a^\dagger$, and hence by the Majoranas, is $\mathrm{End}(\mathcal{F}_-(\mathbb{C}^M))$. Its only invariant subspaces are zero and the whole space.

Put $q = 2r$, $X = \psi_{i_1} \cdots \psi_{i_q}$. Reversing q distinct anticommuting factors takes $q(q-1)/2$ exchanges, so

$$X^* = (-1)^{q(q-1)/2} X = (-1)^r X, \quad X^2 = (-1)^r I, \quad (i^r X)^* = i^r X, \quad (i^r X)^2 = I.$$

For Jordan–Wigner generators on sites $a < b$, the earlier nonidentity matrix anticommutes with the single σ_a^z in the later string, and all other exchanges commute. At one site $\sigma^x\sigma^y = -\sigma^y\sigma^x$; their squares and those of σ^z are I. This proves all Clifford relations for the displayed tensor representation.

Quadratic dynamics and thermal Wick factorization. The identity

$$[\psi_i\psi_j, \psi_k] = 2\delta_{jk}\psi_i - 2\delta_{ik}\psi_j$$

follows by moving ψ_k through two factors. Summing against $iA_{ij}/4$ gives $[H_2, \psi_k] = -i\sum_j A_{kj}\psi_j$. Since $A^T = -A$, e^{tA} is orthogonal, and differentiation verifies both the Heisenberg solution and preservation of the CAR. In the block basis $-i\varepsilon_a\chi_{2a-1}\chi_{2a}/2 = \varepsilon_a(n_a - 1/2)$; summing $e^{-\beta\varepsilon_a(n_a-1/2)}$ over each $n_a \in \{0, 1\}$ gives the product partition function. Degeneracies allow other eigenvectors, but the occupation states always supply an eigenbasis.

Here is a trace proof of Wick's theorem that also covers zero block energies. Write $\langle\cdot\rangle$ for the quadratic Gibbs state. Take $X = d_a$ or $d_a^\dagger$, with $\rho X = rX\rho$ and $r = e^{\beta\varepsilon_a}$ or $e^{-\beta\varepsilon_a}$, respectively. Let $X_2, \ldots, X_{2m}$ be arbitrary linear combinations of the $d_b, d_b^\dagger$. Their anticommutators with X are scalars. Moving X successively to the right gives

$$XX_2 \cdots X_{2m} = \sum_{j=2}^{2m} (-1)^j \{X, X_j\} X_2 \cdots \widehat{X}_j \cdots X_{2m} - X_2 \cdots X_{2m} X.$$

Cyclicity implies $\langle X_2 \cdots X_{2m} X\rangle = r^{-1}\langle XX_2 \cdots X_{2m}\rangle$. The same calculation for two factors gives $(1 + r^{-1})\langle XX_j\rangle = \{X, X_j\}$. Therefore

$$\langle XX_2 \cdots X_{2m}\rangle = \sum_{j=2}^{2m} (-1)^j \langle XX_j\rangle\langle X_2 \cdots \widehat{X}_j \cdots X_{2m}\rangle.$$

Linearity in X and induction give the signed pairing formula. Odd products have zero expectation by parity. Quadratic time evolution preserves linear combinations, so the argument applies to unequal-time insertions, with time ordering supplying its permutation sign.

Pfaffians and ordinary Gaussians. In the Grassmann exponential, only $(\theta^T A\theta/2)^m/m!$ contributes to top degree. Its coefficient is exactly the permutation formula for $\mathrm{Pf}(A)$. A linear change of generators also gives $\mathrm{Pf}(B^T AB) = \det(B)\,\mathrm{Pf}(A)$. For real skew-symmetric A, orthogonal block reduction to matrices $\left(\begin{smallmatrix} 0 & a_j \\ -a_j & 0 \end{smallmatrix}\right)$ then gives $\mathrm{Pf}(A)^2 = \prod_j a_j^2 = \det A$. Both sides are polynomials in the independent matrix entries, so their identity on all real entries proves it for complex skew-symmetric matrices as well. Singular matrices are included.

For real positive-definite symmetric A, orthogonal diagonalization and $y = A^{1/2}x$ reduce the ordinary Gaussian integral to $\det(A)^{-1/2}\prod_j \int_{\mathbb{R}} e^{-y_j^2/2}dy_j$. Squaring the one-dimensional integral and using polar coordinates gives $\int_{\mathbb{R}} e^{-y^2/2}dy = \sqrt{2\pi}$, proving the stated formula.

A.7. Thermal trace calculations

Thermodynamic derivatives and the zero-temperature limit. For a fixed finite matrix, differentiating $Z = \sum_a e^{-\beta E_a}$ gives

$$U = -\partial_\beta \log Z, \quad \partial_\beta U = -\big(\langle H^2\rangle - \langle H\rangle^2\big), \quad S = \log Z + \beta U, \quad C_V = \beta^2 \,\mathrm{Var}_\rho(H) \geq 0.$$

The entropy formula follows from $\log\rho = -\beta H - \log Z$. It also gives $S = -\partial_T F$. If the distinct ground energy has multiplicity d_0, factor $e^{-\beta\mathcal{E}_0}$ out of Z. Every other Boltzmann weight tends to zero relative to it, so ρ tends to the normalized ground projection and $S \to \log d_0$, using continuity of $-x\log x$ at zero. This is a fixed-N statement.

There is a useful uniform energy estimate needing no gap assumption. With $D = 2^{N/2}$,

$$e^{-\beta\mathcal{E}_0} \leq Z \leq De^{-\beta\mathcal{E}_0}, \qquad \frac{\mathcal{E}_0}{N} - \frac{\log 2}{2\beta} \leq f_{N,g}(\beta) \leq \frac{\mathcal{E}_0}{N}.$$

Thus convergence of the ground-energy density implies convergence of the free-energy density along any $\beta_N \to \infty$. This does not imply ground-state dominance of Z in relative error or convergence of microscopic spacings.

Thermal boundary conditions and Matsubara frequencies. For $0 < \tau < \beta$, trace cyclicity gives

$$\langle A(\tau-\beta)B\rangle_\beta = \langle BA(\tau)\rangle_\beta.$$

Applying the minus sign in fermionic time ordering to the left-hand insertion proves $G_{AB}(\tau-\beta) = -G_{AB}(\tau)$. This is the fermionic form of the Kubo–Martin–Schwinger (KMS) thermal boundary condition. Fourier modes satisfy $e^{-i\omega\beta} = -1$, hence $\omega = (2n+1)\pi/\beta$. For $H = 0$, $G_0(\tau) = 1$ on $(0,\beta)$, so $G_0(i\omega_n) = (e^{i\omega_n\beta}-1)/(i\omega_n) = -2/(i\omega_n)$. Equivalently, its antiperiodic extension has jump two and $\frac{1}{2}\partial_\tau G_0$ is the identity delta kernel.

Lehmann formulas and positivity. For a self-adjoint ψ, insertion of an energy eigenbasis gives

$$G(\tau) = \frac{1}{Z}\sum_{n,m} e^{-\beta E_n}e^{-\tau\omega_{mn}}|\psi_{nm}|^2, \qquad \omega_{mn} = E_m - E_n, \quad 0 < \tau < \beta.$$

Consequently $G(\tau) = \int e^{-\tau\omega}(1+e^{-\beta\omega})^{-1}\,d\nu_\psi(\omega)$. Since $e^{i\omega_n\beta} = -1$,

$$\int_0^\beta e^{(i\omega_n-\omega)\tau}d\tau = -\frac{1+e^{-\beta\omega}}{i\omega_n-\omega},$$

which proves the Euclidean formula. Expanding the real-time anticommutator gives $\langle\{\psi(t),\psi\}\rangle = \int e^{-i\omega t}d\nu_\psi(\omega)$. For $\operatorname{Im} z > 0$, $-i\int_0^\infty e^{i(z-\omega)t}dt = (z-\omega)^{-1}$, proving the retarded formula and its sign relative to G.

All weights in ν_ψ are nonnegative. Its total mass is $2\operatorname{Tr}\big(\rho\psi^2\big) = 2$, and interchanging m, n shows it is even. In particular

$$\operatorname{Im} G^R(x+iy) = -\int \frac{y}{(x-\omega)^2+y^2}\, d\nu_\psi(\omega) \le 0, \qquad y > 0.$$

At finite N this function has only real spectral poles. A continuous spectral density or branch singularity requires a limiting or smoothing procedure; it is not a singularity of a finite-matrix resolvent.

Pressure differentiation and Gaussian concentration. For matrices H and a variation V, the Duhamel identity is

$$\frac{d}{ds}e^{-\beta(H+sV)}\bigg|_{s=0} = -\int_0^\beta e^{-(\beta-u)H}Ve^{-uH}\,du.$$

It follows by differentiating the evolution equation for the exponential, or its convergent power series. Taking traces makes every integrand $\operatorname{Tr}\big(Ve^{-\beta H}\big)$. Dividing by NZ proves the gradient formula in Proposition 1.4 without assuming $[H,V]=0$.

For completeness, the Gaussian functional inequalities used there are $\operatorname{Var}_\gamma f \le \mathbb{E}_\gamma|\nabla f|^2$ and

$$\operatorname{Ent}_\gamma(e^u) \le \tfrac{1}{2}\mathbb{E}_\gamma(e^u|\nabla u|^2), \quad \operatorname{Ent}_\gamma Y = \mathbb{E}_\gamma(Y\log Y) - \mathbb{E}_\gamma Y \log \mathbb{E}_\gamma Y.$$

They admit short semigroup proofs. Define $P_tf(x) = \mathbb{E}f(e^{-t}x + \sqrt{1-e^{-2t}}Z)$ with Z standard Gaussian. Its invariant law is γ, its generator is $\Delta - x\cdot\nabla$, and integration by parts gives

$$\operatorname{Var}_\gamma f = 2\int_0^\infty \mathbb{E}_\gamma|\nabla P_tf|^2dt \le 2\int_0^\infty e^{-2t}\mathbb{E}_\gamma|\nabla f|^2dt,$$
$$\operatorname{Ent}_\gamma Y = \int_0^\infty \mathbb{E}_\gamma\frac{|\nabla P_tY|^2}{P_tY}dt \le \int_0^\infty e^{-2t}\mathbb{E}_\gamma\frac{|\nabla Y|^2}{Y}dt.$$

Here $\nabla P_t = e^{-t}P_t\nabla$; Jensen proves the first estimate, and weighted Cauchy–Schwarz proves the second. These identities follow by differentiating $\mathbb{E}(P_tf)^2$ and $\mathbb{E}(P_tY)\log(P_tY)$ and integrating to equilibrium. Smooth truncation extends them from bounded positive smooth functions to the exponential of a Lipschitz function, using Gaussian exponential integrability.

If f is L-Lipschitz and $\varphi(s) = \log\mathbb{E}e^{s(f-\mathbb{E}f)}$, the entropy inequality yields $s\varphi'(s)-\varphi(s) \le s^2L^2/2$. Integrating $(\varphi(s)/s)' \le L^2/2$ from zero gives $\varphi(s) \le s^2L^2/2$. Markov's inequality with $s = t/L^2$ proves $\mathbb{P}(f-\mathbb{E}f \ge t) \le e^{-t^2/(2L^2)}$; apply it to $-f$ for the other tail. This proves the concentration bounds used in the text.

Replica differentiation and Taylor coefficients. At fixed N, $\|H\| \le \sigma_N\sum_I|g_I|$ and $|\log Z| \le \log D + \beta\|H\|$. Gaussian integrability therefore gives $\mathbb{E}e^{a|\log Z|} < \infty$ for every finite a. For real n near zero,

$$\left|\frac{Z^n-1}{n}\right| \le |\log Z|e^{|n||\log Z|}.$$

Dominated convergence proves (5.1). The same norm bound justifies termwise disorder expectation of the exponential series for Z. At $\beta = 0$, differentiating $N^{-1} \log Z$ twice gives $N^{-1}(\tau_N(H^2) - \tau_N(H)^2)$. Clifford orthogonality then gives the stated random coefficient. Finally $\mathbb{E}g^2 = 1$, $\mathbb{E}g^4 = 3$ yield $\mathrm{Var}(g^2) = 2$ and its variance formula; the independent identically distributed central limit theorem applies to the $m_N \to \infty$ summands. Details omitted: a proof of that standard central limit theorem.

The Schwarzian Laplace transform. Completing the square gives

$$J(a) = \int_0^\infty e^{-\beta x^2} \cosh(ax) dx = \frac{\sqrt{\pi}}{2\sqrt{\beta}} e^{a^2/(4\beta)}.$$

Differentiation under the Gaussian integral is justified by domination. With $\varepsilon = x^2$, the desired integral is $2J'(a) = a\sqrt{\pi}\, e^{a^2/(4\beta)}/(2\beta^{3/2})$. Multiplication by $C/(2\pi^2)$ and $a = 2\pi\sqrt{2C}$ gives exactly (2.12). Its full smooth entropy and heat capacity are

$$\begin{aligned} S_{\text{smooth}} &= S_0 + 4\pi^2 CT + \tfrac{3}{2}\log(CT) + \tfrac{3}{2} - \tfrac{1}{2}\log(2\pi), \\ C_{V,\text{smooth}} &= 4\pi^2 CT + \tfrac{3}{2}. \end{aligned}$$

These identities assume E_0, S_0, C independent of T. Their extrapolation to $T = 0$ does not describe a finite matrix. Near the smooth edge, $\sinh x = x + O(x^3)$ gives

$$\rho_{\text{smooth}}(E_0 + \varepsilon) = e^{S_0} \frac{\sqrt{2}}{\pi} C^{3/2} \varepsilon^{1/2} (1 + O(C\varepsilon)).$$

Integrating gives coefficient $2\sqrt{2}/(3\pi)$ in the cumulative law. Equating this predicted count to one proves the conditional scale estimates in the text. In the general power-law version one needs $\alpha > -1$ so that the edge count is finite; then $\int_0^\varepsilon u^\alpha du = \varepsilon^{\alpha+1}/(\alpha+1)$.

A.8. Conformal distributions and coordinate changes

The normalization of the power-law solution. Set $a = 2/q \in (0,1)$ and $f_a(t) = \mathrm{sgn}(t)|t|^{-a}$. The regulated Gamma integral is

$$\int_0^\infty e^{-(\epsilon - i\omega)t} t^{-a} dt = \Gamma(1-a)(\epsilon - i\omega)^{a-1}, \quad \epsilon > 0.$$

Subtract the negative half-line contribution and let $\epsilon \downarrow 0$. This gives $\widehat{f}_a = 2i\Gamma(1-a)\cos(\pi a/2)\,\mathrm{sgn}(\omega)|\omega|^{a-1}$.

The self-energy exponent $2 - a$ exceeds one, so its naive integral at zero diverges. Define the odd homogeneous distribution f_{2-a} by the symmetric principal value

$$\langle f_{2-a}, \varphi \rangle = \int_0^\infty t^{a-2}(\varphi(t) - \varphi(-t)) dt$$

for Schwartz φ. The difference vanishes linearly at zero, making this integral finite. Equivalently $f_{2-a} = (a-1)^{-1}\partial_t |t|^{a-1}$. The even Gamma transform of $|t|^{a-1}$ is $2\Gamma(a)\cos(\pi a/2)|\omega|^{-a}$, so

$$\widehat{f}_{2-a} = -2i\Gamma(a-1)\cos(\pi a/2)\,\mathrm{sgn}(\omega)|\omega|^{1-a}.$$

The product of these locally integrable Fourier functions is constant off zero and defines a constant distribution. The Gamma reflection identity gives

$$\widehat{f_a}\widehat{f_{2-a}} = 4\Gamma(1-a)\Gamma(a-1)\cos^2(\pi a/2) = -\frac{2\pi}{1-a}\cot(\pi a/2).$$

With $G = bf_a$ and the homogeneous extension $\Sigma = b^{q-1}f_{2-a}$, the Fourier form of $\Sigma \circ G = -I$ is therefore equivalent to

$$b^q = \frac{1-a}{2\pi}\tan(\pi a/2).$$

For $a = 1/2$ this is $b^4 = 1/(4\pi)$. This specifies convolution through Fourier multiplication; an unregularized time integral would not justify the computation. Also $|\widehat{\Sigma}|/|\omega|$ is proportional to $|\omega|^{-a} \to \infty$, verifying the self-consistency of dropping the kinetic term within this ansatz. Details omitted: the standard Gamma reflection identity.

Covariance and the thermal circle. Let $f' > 0$ and $G_f(1,2) = (f_1'f_2')^{\Delta}G(f_1,f_2)$, $\Sigma_f(1,2) = (f_1'f_2')^{1-\Delta}\Sigma(f_1,f_2)$. The power relation is preserved because $\Delta(q-1) = 1-\Delta$. Changing variables $v = f(\tau_3)$ in the convolution gives

$$(\Sigma_f \circ G_f)_{12} = -(f_1')^{1-\Delta}(f_2')^{\Delta}\delta(f_1 - f_2) = -\delta(\tau_1 - \tau_2).$$

The last equality uses $\delta(f_1 - f_2) = \delta(\tau_1 - \tau_2)/f_2'$ and the equality of the derivatives on the diagonal. This is valid for compatible distributional kernels, or first away from singularities followed by their covariant distributional extension.

In a projective chart put $f(\tau) = \tan(\pi\tau/\beta)$. The identity $\tan x - \tan y = \sin(x-y)/(\cos x \cos y)$ gives

$$\frac{[f'(\tau_1)f'(\tau_2)]^{\Delta}}{|f(\tau_1) - f(\tau_2)|^{2\Delta}} = \left|\frac{\pi/\beta}{\sin(\pi\tau_{12}/\beta)}\right|^{2\Delta}.$$

Orientation fixes the sign within a chart; the fermionic antiperiodic extension fixes it across the chart boundary. This yields (2.6). For $M(x) = (ax+b)/(cx+d)$, $ad - bc = 1$, the identities $M'(x) = (cx+d)^{-2}$ and $M(x) - M(y) = (x-y)/((cx+d)(cy+d))$ prove projective invariance locally. On the circle the global statement uses the same fermionic boundary condition.

Schwarzian chain rule and expansion. Write $\ell_f = f''/f'$. Then $\{f,t\} = \ell_f' - \frac{1}{2}\ell_f^2$ and $\ell_{F\circ f} = (\ell_F \circ f)f' + \ell_f$. Differentiating and canceling the mixed terms proves

$$\{F \circ f, t\} = (f')^2\{F, f(t)\} + \{f,t\}.$$

For a fractional linear M, $\ell_M = -2c/(cx+d)$, so $\ell_M' - \ell_M^2/2 = 0$. This supplies all cancellations in the projective-invariance calculation above. For $F(x) = \tan(ax)$, $\ell_F = 2a\tan(ax)$, hence $\{F,x\} = 2a^2$. The chain rule gives

$$\{\tan(\pi f/\beta), \tau\} = \{f,\tau\} + 2\pi^2(f')^2/\beta^2.$$

For a smooth circle lift, f''/f' is periodic, so its derivative integrates to zero. Rescaling $\tau = \beta u$, $f = \beta g$ gives (2.10) and the integrated formula for $\mathcal{A}[g]$. With $g = u + \varepsilon h$ and $|\varepsilon|\|h'\|_\infty < 1$, expand $(g''/g')^2 = \varepsilon^2(h'')^2 + O_h(\varepsilon^3)$ and $(g')^2 = 1 + 2\varepsilon h' + \varepsilon^2(h')^2$. Since $\int_0^1 h' = 0$, the claimed quadratic expansion follows.

A.9. What the formal linearizations prove

Disorder coefficient for general even interaction order. With a finite time regulator, put $x_i = \psi_i(\tau_1)\psi_i(\tau_2)$ in a Grassmann algebra. The x_i commute and $x_i^2 = 0$, so $(\sum_i x_i)^q = q! \sum_{i_1<\cdots<i_q} x_{i_1}\cdots x_{i_q}$. Grouping the two sets of q Grassmann factors contributes $(-1)^{q(q-1)/2}$; their phases contribute $i^q = (-1)^{q/2}$. Their product is $+1$ for even q. Gaussian averaging thus gives exactly, at this algebraic level,

$$\frac{\sigma_N^2}{2q!}\int d\tau_1 d\tau_2 \left(\sum_i x_i\right)^q = \frac{N}{2q}\int G_{12}^q \, d\tau_1 d\tau_2.$$

The Gaussian identity extends to these nilpotent even elements by its finite Taylor expansion. Passage to a continuum functional integral still needs a regulator and contour; details omitted.

Hessian scaling and the ladder kernel. For an ordinary finite-dimensional integral with a nondegenerate stable minimum x_*, write $x = x_* + N^{-1/2}\xi$. Then

$$NI(x) = NI(x_*) + \tfrac{1}{2}\xi^T H \xi + O(N^{-1/2}|\xi|^3), \quad H = I''(x_*).$$

Under hypotheses permitting Laplace asymptotics, covariance is $N^{-1}H^{-1}$ to leading order. In a direction with restoring action coefficient C/β, the analogous Gaussian width is $(\beta/C)^{1/2}$. Neither conclusion is uniform when the Hessian develops zero eigenvalues.

The sign and factor of the ladder kernel follow directly by differentiating the formal Dyson relation. Since $\delta A = -\delta\Sigma$, $\delta G = G \circ \delta\Sigma \circ G$, and $\delta\Sigma_{34} = (q-1)G_{34}^{q-2}\delta G_{34}$. Antisymmetry gives $G_{42} = -G_{24}$, hence

$$\delta G_{12} = -\int d\tau_3 d\tau_4 \, (q-1) G_{13} G_{24} G_{34}^{q-2} \delta G_{34}.$$

This is the kernel in the text in the full ordered-pair convention. With an external source the corresponding inhomogeneous equation is $(I - K)\delta G = \text{source}$. Differentiating a reparametrized family of IR solutions gives $K\delta G = \delta G$, explaining why a reparametrization direction obstructs inversion. Its representation-theoretic label $h = 2$ requires the conformal spectral analysis omitted earlier.

Quasiparticle residue and lifetime. Suppose $\varepsilon_{k_F} + \operatorname{Re}\Sigma^R(0, k_F) = 0$, the real part is differentiable, and damping is sufficiently small. Taylor expansion gives

$$\omega - \varepsilon_k - \Sigma^R(\omega, k) = a\omega - bk_\perp - i \operatorname{Im}\Sigma^R(\omega, k) + o(|\omega| + |k_\perp|),$$

where $a = 1 - \partial_\omega \operatorname{Re}\Sigma^R(0, k_F)$ and $b = \partial_{k_\perp}(\varepsilon_k + \operatorname{Re}\Sigma^R(0, k))|_{k_F}$. For $a > 0$, division by a yields $Z = a^{-1}$, $v_F^* = b/a$, and $\Gamma = -Z \operatorname{Im}\Sigma^R \geq 0$ when the retarded damping has the usual sign. A pole $(\omega - E + i\Gamma)^{-1}$ Fourier transforms for positive time to $-ie^{-iEt-\Gamma t}$. The amplitude lifetime is Γ^{-1}, and many oscillations before damping means $\Gamma/|E| \to 0$; a probability decay time differs by a factor two.

For the conditional scaling $G^R(\omega) \asymp |\omega|^{a-1}$ with $0 < a < 1$, one has $\omega G^R(\omega) \to 0$. At zero momentum displacement a quasiparticle pole with residue $Z_* > 0$ and $\Gamma = o(|\omega|)$ would instead give $\omega G^R(\omega) \to Z_*$. Thus this specific scaling excludes such a pole if the remainder is less singular. Self-energy dominance alone, without these analytic and remainder assumptions, is not a general classification theorem.

A.10. Antiunitary symmetries and the late-time average

The quartic symmetry table. In the Jordan–Wigner basis, let $\mathcal{K}$ be entrywise complex conjugation. Odd-numbered Majoranas are real and even-numbered ones imaginary. For $M = N/2$, set $U = \psi_1\psi_3\cdots\psi_{2M-1}$ and $\mathcal{T} = U\mathcal{K}$. Conjugation by U multiplies a Majorana in this product by $(-1)^{M-1}$ and one outside it by $(-1)^M$. Therefore

$$\mathcal{T}\psi_i\mathcal{T}^{-1} = (-1)^{M-1}\psi_i, \quad \mathcal{T}^2 = U\overline{U} = U^2 = (-1)^{M(M-1)/2}I.$$

Quartic terms have a real phase $i^2 = -1$ and four Majoranas, so $\mathcal{T}H_4\mathcal{T}^{-1} = H_4$. Also $P = (-1)^F = \prod_a i\psi_{2a-1}\psi_{2a}$, so $\mathcal{T}P\mathcal{T}^{-1} = (-1)^M P$ by conjugate linearity. If M is odd it exchanges parity sectors. If $M \equiv 0 \pmod 4$ it preserves a sector and squares to $+I$; if $M \equiv 2 \pmod 4$ it preserves a sector and squares to $-I$. These are exactly the cases in the table.

For an antiunitary with $\mathcal{T}^2 = -I$, put $a = \langle v, \mathcal{T}v\rangle$ with the inner product conjugate-linear in its first argument. The antiunitary identity gives $\langle \mathcal{T}v, \mathcal{T}^2 v\rangle = \overline{a}$, whereas $\mathcal{T}^2 v = -v$ makes the same quantity $-\overline{a}$. Thus $a = 0$; commuting with H gives the energy degeneracy. A square-$+I$ antiunitary supplies a real structure, and a square-$-I$ one a quaternionic structure. These algebraic constraints motivate GOE and GSE comparison; they do not make the SYK distribution Gaussian in all matrix entries.

The plateau as a time average. At fixed sample and fixed β, the sum defining $|Z(\beta + it)|^2$ is finite. For $\Delta E \neq 0$,

$$\frac{1}{T}\int_0^T e^{-it\Delta E}dt = \frac{1 - e^{-iT\Delta E}}{iT\Delta E} \longrightarrow 0.$$

Consequently its Cesàro time average is $\sum_E d_E^2 e^{-2\beta E}$, or $Z(2\beta)$ when all eigenvalues being counted are simple. This does not assert a pointwise $t \to \infty$ limit, which generally fails for a finite quasiperiodic sum. Disorder expectation can be passed through this time-average limit at fixed N by the bound $|Z(\beta + it)|^2 \le D^2 e^{2\beta\|H\|}$ and Gaussian integrability.

A.11. Geometric computations

Curvature, isometries, and the mass equation. For $g = L^2 z^{-2}(-dt^2 + dz^2)$, the nonzero Christoffel symbols are

$$\Gamma^t_{tz} = \Gamma^t_{zt} = -z^{-1}, \quad \Gamma^z_{tt} = -z^{-1}, \quad \Gamma^z_{zz} = -z^{-1}.$$

Substitution in $R_{\mu\nu} = \partial_\lambda\Gamma^\lambda_{\mu\nu} - \partial_\nu\Gamma^\lambda_{\mu\lambda} + \Gamma^\lambda_{\lambda\rho}\Gamma^\rho_{\mu\nu} - \Gamma^\lambda_{\nu\rho}\Gamma^\rho_{\mu\lambda}$ gives $R_{tt} = z^{-2}$, $R_{zz} = -z^{-2}$, $R_{tz} = 0$, hence $R_{\mu\nu} = -g_{\mu\nu}/L^2$ and $R = -2/L^2$.

The vector fields $P = \partial_t$, $D = t\partial_t + z\partial_z$, and $K = (t^2 + z^2)\partial_t + 2tz\partial_z$ satisfy $\mathcal{L}_X g = 0$. This follows by substituting them into $(\mathcal{L}_X g)_{\mu\nu} = X^\lambda\partial_\lambda g_{\mu\nu} + g_{\lambda\nu}\partial_\mu X^\lambda + g_{\mu\lambda}\partial_\nu X^\lambda$. Their brackets are $[D, P] = -P$, $[D, K] = K$, $[P, K] = 2D$, the Lie algebra $\mathfrak{sl}_2(\mathbb{R})$. At $z = 0$ they act as $\partial_t, t\partial_t, t^2\partial_t$, generating projective time transformations. This establishes the local symmetry statement without identifying global covers.

Since $|\det g|^{1/2} = L^2/z^2$, the scalar wave operator is $\Box_g = L^{-2}z^2(-\partial_t^2 + \partial_z^2)$. For a boundary behavior $\phi(t, z) = z^h a(t)$,

$$(\Box_g - m^2)\phi = L^{-2}[h(h-1) - m^2L^2]z^h a(t) - L^{-2}z^{h+2}a''(t).$$

Vanishing of the leading coefficient proves $m^2L^2 = h(h-1)$, with roots $h = (1 \pm \sqrt{1+4m^2L^2})/2$. Which root is allowed is a boundary-condition question. Details omitted: the bulk stability and quantization theorems.

The topological term and dilaton constraint. For a compact oriented Euclidean surface with smooth boundary, Gauss–Bonnet states

$$\int_M \sqrt{g} R \, d^2x + 2\int_{\partial M} K \, ds = 4\pi\chi(M),$$

with consistent outward-normal convention for K. Here $\chi(M)$ is the Euler characteristic and K the geodesic or extrinsic curvature of the boundary. Thus the constant-dilaton term including its boundary partner is $-\Phi_0\chi(M)/(4G)$ and has no local metric variation at fixed topology. In two dimensions $R_{\mu\nu} = \frac{1}{2}Rg_{\mu\nu}$ also makes the Einstein tensor identically zero. For compactly supported arbitrary $\delta\Phi$, varying the second term of (4.4) gives $-(16\pi G)^{-1}\int\sqrt{g}\delta\Phi(R + 2/L^2)$, so its Euler–Lagrange equation is precisely the stated curvature constraint. Details omitted: a proof of Gauss–Bonnet and the full classification of gravitational degrees of freedom.

The regulated JT boundary action. Here use Euclidean $g = L^2z^{-2}(dt^2 + dz^2)$ and a boundary curve $(t(u), z(u))$ with $t' > 0$. Fix $ds = L\,du/\epsilon$ and boundary dilaton $\Phi_b = \phi_r/\epsilon$, with constant $\phi_r > 0$. The induced metric condition is $t'^2 + z'^2 = z^2/\epsilon^2$. For a fixed smooth t it gives

$$z = \epsilon t' + \frac{\epsilon^3}{2}\frac{t''^2}{t'} + O(\epsilon^5).$$

Indeed substitution equates the order-ϵ^2 coefficients to $2t'(z - \epsilon t')/\epsilon^3 = t''^2$. The outward normal points toward smaller z. Directly from the unit normal, or the conformal change of geodesic curvature, one obtains

$$K = \frac{1}{L}\left(\frac{t'}{\sqrt{t'^2+z'^2}} + \frac{z(t'z'' - z't'')}{(t'^2+z'^2)^{3/2}}\right) = \frac{1}{L}\left[1 + \epsilon^2\left(\frac{t'''}{t'} - \frac{3}{2}\frac{t''^2}{t'^2}\right) + O(\epsilon^4)\right].$$

For the last equality, the first fraction is $1 - \epsilon^2 t''^2/(2t'^2) + O(\epsilon^4)$ and the second is $\epsilon^2(t'''/t' - t''^2/t'^2) + O(\epsilon^4)$. The renormalized dynamical boundary term is consequently

$$-\frac{1}{8\pi G}\int ds\, \Phi_b(K - L^{-1}) = -\frac{\phi_r}{8\pi G}\int \{t, u\}\, du + O(\epsilon^2).$$

The bulk dilaton term vanishes on the curvature constraint. Setting $t = \tan(\pi f/\beta)$ gives (4.5) with $C = \phi_r/(8\pi G)$ in these boundary conventions, apart from the separate topological term. The computation is local in projective charts, with periodic data imposed on the thermal boundary. It assumes a smooth family with derivatives bounded as the cutoff tends to zero.

Surface gravity and boosts. A nonextremal near-horizon metric has local Rindler form $ds^2 \simeq -\kappa^2r^2dt^2 + dr^2$. After $t = -i\tau$, smoothness at $r = 0$ requires $\kappa\tau$ to have period 2π, hence $\beta\kappa = 2\pi$. Minkowski null coordinates can be chosen $X^+ = re^{\kappa t}$ and $X^- = -re^{-\kappa t}$. A time translation by δt multiplies them by $e^{\pm\kappa\delta t}$, proving the relative boost factor. Identifying that boost with an OTOC correction still needs the dynamical scattering argument omitted in the main text.

A.12. Comparison of conventions

Different papers normalize Majorana fermions either by $\{\psi_i, \psi_j\} = \delta_{ij}$ or by $\{\psi_i, \psi_j\} = 2\delta_{ij}$. The convention in these notes is the second one, so $\psi_i^2 = I$. If $\chi_i \coloneqq \psi_i/\sqrt{2}$, then $\{\chi_i, \chi_j\} = \delta_{ij} I$ and

$$i^{q/2}\psi_{i_1}\cdots\psi_{i_q} = 2^{q/2} i^{q/2}\chi_{i_1}\cdots\chi_{i_q}.$$

Thus the same operator H_q has coupling variance $2^q(q-1)!/N^{q-1}$ when written in the χ convention. Correspondingly $G_\psi = 2G_\chi$, while the equations used in these notes are

$$G_\psi^{-1}(i\omega) = -\frac{i\omega}{2} - \Sigma_\psi(i\omega), \qquad \Sigma_\psi(\tau) = G_\psi(\tau)^{q-1}.$$

Likewise some authors write $G(\tau) = -\langle T_\tau \chi(\tau)\chi(0)\rangle$ with an explicit minus sign, while others absorb it into conventions. The universal structural statements are unaffected:

$$\Sigma \propto J^2 G^{q-1}, \qquad \Delta = 1/q, \qquad I_{\text{soft}} \propto \frac{N}{J} I_{\text{Sch}}, \qquad \lambda_L \to 2\pi/\beta.$$

When comparing formulas between papers, one should first align these conventions.

B. Terminology

Physics term	Mathematical translation
Majorana fermion	Self-adjoint Clifford generator ψ_i with $\{\psi_i, \psi_j\} = 2\delta_{ij} I$.
Quenched disorder	Random Hamiltonian sampled once, then thermodynamics computed at fixed sample before averaging $\mathbb{E} \log Z$.
Self-energy	Kernel Σ entering the nonlinear resolvent identity $G^{-1} = G_0^{-1} - \Sigma$.
Large N	Asymptotic regime in which singlet fluctuations are suppressed and melonic diagrams dominate.
Melon	Recursively generated leading graph maximizing powers of the flavor index N.
Infrared	Frequency regime $\|\omega\| \ll 1$, or time regime $\|\tau\| \gg 1$, with the effective large-N coupling scale set to one.
Emergent conformal symmetry	Approximate scale/reparametrization covariance of the low-energy Schwinger-Dyson equations after dropping the kinetic term.
Soft mode	Nearly zero-cost reparametrization fluctuation $f \in \mathrm{Diff}\,/PSL(2,\mathbb{R})$.
Schwarzian	Effective action for the soft mode, built from $\{f, \tau\}$.
Quasiparticle	Pole-like excitation with width $\Gamma = o(\|E\|)$.
Non-Fermi liquid	Gapless fermionic state lacking the Landau quasiparticle pole structure.
OTOC	Four-point function probing growth of noncommutativity and operator complexity.
Maximal chaos	Exponential OTOC growth with $\lambda_L = 2\pi/\beta$, saturating the chaos bound.
AdS_2	Constant negative curvature two-dimensional spacetime with isometry group $SL(2,\mathbb{R})$.
JT gravity	Two-dimensional dilaton gravity whose boundary mode is governed by a Schwarzian action.
Holography	Reconstruction of bulk gravitational observables from lower-dimensional quantum correlation data in an appropriate limit.

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